



ISE 430/530: Manufacturing Systems and Facilities Design

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Facilities Planning Process

- Define products to be manufactured and/or assembled
- Specify required manufacturing and/or assembly processes and related activities
- Determine interrelationships among all activities
- Generate alternative facilities plans
- Evaluate the alternative facilities plans
- Select the preferred facilities plan
- Implement the facilities plan
- Maintain and adapt the facilities plan
- Update products to be manufactured and/or assembled and redefine the objective of the facility

Overview

Route sheet, assembly chart, operations process chart, precedence diagram

Determining the Production Requirements

- How many pieces are needed as inputs and outputs to each process, consider rework (if appropriate)
- Reject allowance (small volume)
- Number of required machines
- Machine assignment

Reject Allowance Problem

1. When producing small batches, the use of average values is less appropriate
2. Concerns
 - Cost, revenues, and probabilities associated with making good and bad products
3. Objective
 - Determine the number of additional units to allow when scheduling low-volume production where rejects occur randomly
4. For instance, consider a foundry that produces small numbers of custom-designed castings.
5. If the foundry has only one chance to produce the number of castings required, then the probability of a casting being good should be considered when determining the batch size to be produced.

Reject Allowance Problem

Determining the number of additional units to allow when scheduling low-volume production where rejects randomly occur is called the reject allowance problem.

x = the number of good units produced

$p(x)$ = probability of producing x good units

Q = quantity of units to produce

$C(Q, x)$ = cost of producing Q units, of which x are good units

$R(Q, x)$ = revenue from producing Q units, of which x are good units

$P(Q, x)$ = profit from producing Q units, of which x are good units

$$= R(Q, x) - C(Q, x)$$

$E[P(Q)]$ = expected profit from producing Q units

$$= \sum_{x=0}^Q P(Q, x) p(x)$$

The expected profit from producing Q units can be determined as follows:

$$E[P(Q)] = \sum_{x=0}^Q \{R(Q, x) - C(Q, x)\} p(x) \quad (2.4)$$

Reject Allowance Problem – Example:

An order for 20 custom-designed castings has been received. Each casting costs \$1100 to produce and has a recycle value of \$200 if unsold. The customer will pay \$2500 per casting for exactly 20 castings —no more, no less!. Based on historical data, Table 2.6 provides the probability distributions.

1. How many castings should be produced to maximize expected profit?

2. What is the probability of losing money at this production level?

Table 2.6 *Probability Distributions for the Number of Good Castings (x) out of Q*

# Good Castings (x)	Number of Castings Produced (Q)										
	20	21	22	23	24	25	26	27	28	29	30
12	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
13	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
14	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
15	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00	0.00
16	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00	0.00
17	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00	0.00
18	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00	0.00
19	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00	0.00
20	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00	0.00
21	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05	0.00
22	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05	0.05
23	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05	0.05
24	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05	0.05
25	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10	0.05
26	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10	0.10
27	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15	0.10
28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20	0.15
29	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25	0.20
30	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.25

Reject Allowance Problem – Example:

The revenue and cost functions can be given as follows:

$$R(Q, X) = \begin{cases} 200Q & x < 20 \\ 2500(20) + 200(Q - 20) & 20 \leq x \leq Q \end{cases}$$

$$C(Q, x) = 1,100Q \quad 0 \leq x \leq Q$$

$$P(Q, x) = \begin{cases} -900Q & x < 20 \\ 46,000 - 900Q & 20 \leq x \leq Q \end{cases}$$

Therefore, the expected profit can be given as follows:

$$E[P(Q)] = -\sum_{x=0}^{19} 900Q p(x) + \sum_{x=20}^Q (46,000 - 900Q)p(x)$$

The expected profit expression can be shown to reduce to

$$E[P(Q)] = -900Q + 46,000 \sum_{x=20}^Q p(x)$$

Plug in the numbers in Table 2.6

Table 2.7 Profit from Producing Q Castings, with Exactly x Being Good

# of Good Castings	Number of Castings Produced (Q)										
	20	21	22	23	24	25	26	27	28	29	30
12	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
13	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
14	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
15	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
16	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
17	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
18	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
19	-18,000	-18,900	-19,800	-20,700	-21,600	-22,500	-23,400	-24,300	-25,200	-26,100	-27,000
20	28,000	27,100	26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
21		27,100	26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
22			26,200	25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
23				25,300	24,400	23,500	22,600	21,700	20,800	19,900	19,000
24					24,400	23,500	22,600	21,700	20,800	19,900	19,000
25						23,500	22,600	21,700	20,800	19,900	19,000
26							22,600	21,700	20,800	19,900	19,000
27								21,700	20,800	19,900	19,000
28									20,800	19,900	19,000
29										19,900	19,000
30											19,000

Table 2.8 Expected Profit from Producing Q Castings

Number of Castings Produced (Q)										
20	21	22	23	24	25	26	27	28	29	30
-6,500	1,800	7,800	11,500	15,200	16,600	18,000	19,400	20,800	19,900	19,000

MAXIMUM



Number of Required Machines

Man & machine balancing

Machine (equipment) fraction

- Number of machines of one type required to produce the required volume of products
 - Some machines used for multiple operations
 - Some machines may not be able to perform the required operations fast enough
- Calculation based upon ALL operations the machine will perform

Number of Required Machines

The machine fraction for an operation is determined by dividing the total time required to perform the operation by the time available to complete the operation. The total time required to perform an operation is the product of the standard time for the operation and the number of times the operation is to be performed.

1. Are the parts actually being made to the 0.5 hour per part standard time?
2. Is the machine available when needed during the two-hour period?
3. Are the standard time, the number of parts, and the time the machine takes known with certainty and fixed over time?

$$F = \frac{SQ}{EHR}$$

Diagram illustrating the formula for Machine Fraction (F):

- S : Standard time per unit
- Q : Number of units
- E : Machine Fraction
- H : Performance (% time)
- R : Reliability of machine, expressed as percent uptime
- T : Time available per machine

Number of Required Machines - Example

See Eq. 2.5 and Example 2.5

- Standard machining time of 2.8 minutes per part, 200 units needed in 8-hour shift, milling machine will be operating 80% of time. During operational time, parts are produced at 95% rate

$$F = \frac{SQ}{EHR} = \frac{2.8\text{min} * 200}{480\text{min} * 0.8 * 0.95} = 1.54 \text{ machines needed}$$

Number of Required Machines

Combine requires for identical machine types

Table 2.9 *Total Equipment Requirement Specification Example*

Operation Number	Equipment Fraction	Next Highest Whole Number
109	1.1	2
206	2.3	3
274	0.6	1

It may be seen that a minimum of four and a maximum of six machines are required.

With no further information, a specific recommendation on the number of required machines cannot be made.

How many machines are needed?

Unclear, need more information

Job Shop/Production Line

Cost, Set-ups, Inventory, etc.

Future growth

Overview

Determining the Production Requirements

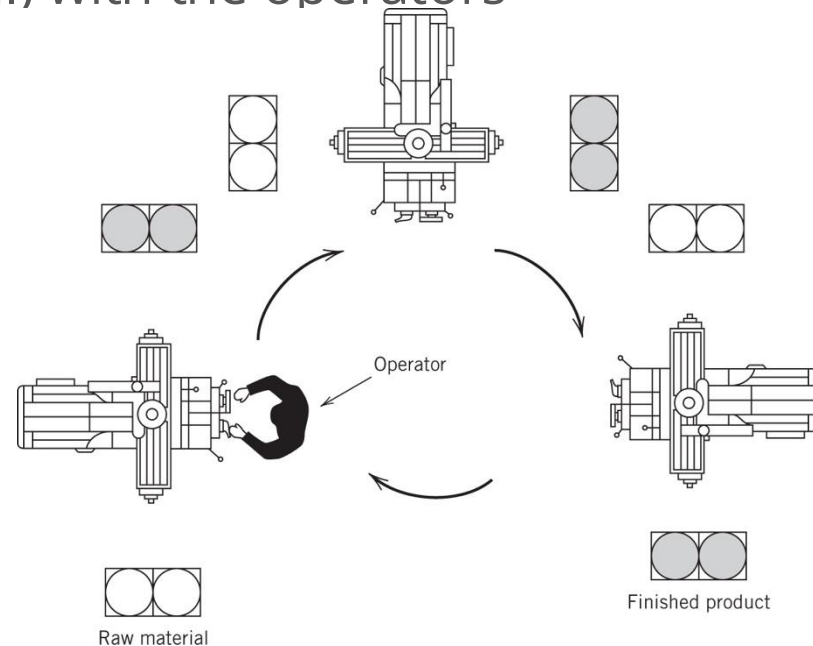
- How many pieces are needed as inputs and outputs to each process, consider rework (if appropriate)
- Reject allowance (small volume)
- Number of required machines
- Machine assignment

Machine Assignment Problems

Considers the assignment of machines (semi-automated) to operators

Human-machine chart (aka. multiple activity chart)

- Relationship of multiple activities and machines (non-identical) with the operators



Time	Operator	Machine 1	Machine 2	Machine 3		
0	L-1	Loaded	Idle	Idle	Transient State (12 min)	
1	T-2	Machining	Idle			
2	L-2		Loaded			
3	T-3		Loaded			
4	L-3					
5	T-1		Machining			
6	Idle					
7	UL-1	Unloaded	Machining	Steady-State Repeating Cycle (9 min)		
8	L-1	Loaded			Idle	
9	I&P-1	Machining	Idle		Idle	
10	T-2		Unloaded			
11	UL-2		Loaded		Idle	
12	L-2		Machining			
13	I&P-2		Unloaded			
14	T-3		Loaded			
15	UL-3		Machining		Machining	
16	L-3	Idle				
17	I&P-3	Machining				
18	T-1	Unloaded	Idle		Machining	
19	UL-1	Loaded				
20	L-1	Machining				
21	I&P-1	Unloaded	Idle		Machining	
22	T-2	Loaded				
23	UL-2	Machining	Machining			
24	L-2		Machining			
25	I&P-2		Machining			
26	T-3		Machining			
27	UL-3	Machining	Machining			
28	L-3		Idle			
29	I&P-3		Machining			
30	T-1	Unloaded	Idle		Machining	
31	UL-1	Loaded				
32	L-1	Machining				
33	I&P-1	Unloaded	Idle		Machining	
34	T-2	Loaded				
35	UL-2	Machining	Machining			
36	L-2		Machining			
37	I&P-2		Machining			
38	T-3		Machining			
39	UL-3	Machining	Machining			
40	L-3		Idle			
41	I&P-3		Machining			
42	T-1	Unloaded	Idle		Machining	
43	UL-1	Loaded				
44	L-1	Machining				
45	I&P-1	Unloaded	Idle		Machining	
46	T-2	Loaded				
47	UL-2	Machining	Machining			
48	L-2		Machining			
49	I&P-2		Machining			
50	T-3		Machining			
51	UL-3	Machining	Machining			
52	L-3		Idle			
53	I&P-3		Machining			
54	T-1	Unloaded	Idle		Machining	
55	UL-1	Loaded				
56	L-1	Machining				
57	I&P-1	Unloaded	Idle		Machining	
58	T-2	Loaded				
59	UL-2	Machining	Machining			
60	L-2		Machining			
61	I&P-2		Machining			
62	T-3		Machining			
63	UL-3	Machining	Machining			
64	L-3		Idle			
65	I&P-3		Machining			
66	T-1	Unloaded	Idle		Machining	
67	UL-1	Loaded				
68	L-1	Machining				
69	I&P-1	Unloaded	Idle		Machining	
70	T-2	Loaded				
71	UL-2	Machining	Machining			
72	L-2		Machining			
73	I&P-2		Machining			
74	T-3		Machining			
75	UL-3	Machining	Machining			
76	L-3		Idle			
77	I&P-3		Machining			
78	T-1	Unloaded	Idle		Machining	
79	UL-1	Loaded				
80	L-1	Machining				
81	I&P-1	Unloaded	Idle		Machining	
82	T-2	Loaded				
83	UL-2	Machining	Machining			
84	L-2		Machining			
85	I&P-2		Machining			
86	T-3		Machining			
87	UL-3	Machining	Machining			
88	L-3		Idle			
89	I&P-3		Machining			
90	T-1	Unloaded	Idle		Machining	
91	UL-1	Loaded				
92	L-1	Machining				
93	I&P-1	Unloaded	Idle		Machining	
94	T-2	Loaded				
95	UL-2	Machining	Machining			
96	L-2		Machining			
97	I&P-2		Machining			
98	T-3		Machining			
99	UL-3	Machining	Machining			
100	L-3		Idle			
101	I&P-3		Machining			
102	T-1	Unloaded	Idle		Machining	
103	UL-1	Loaded				
104	L-1	Machining				
105	I&P-1	Unloaded	Idle		Machining	
106	T-2	Loaded				
107	UL-2	Machining	Machining			
108	L-2		Machining			
109	I&P-2		Machining			
110	T-3		Machining			
111	UL-3	Machining	Machining			
112	L-3		Idle			
113	I&P-3		Machining			
114	T-1	Unloaded	Idle		Machining	
115	UL-1	Loaded				
116	L-1	Machining				
117	I&P-1	Unloaded	Idle		Machining	
118	T-2	Loaded				
119	UL-2	Machining	Machining			
120	L-2		Machining			
121	I&P-2		Machining			
122	T-3		Machining			
123	UL-3	Machining	Machining			
124	L-3		Idle			
125	I&P-3		Machining			
126	T-1	Unloaded	Idle		Machining	
127	UL-1	Loaded				
128	L-1	Machining				
129	I&P-1	Unloaded	Idle		Machining	
130	T-2	Loaded				
131	UL-2	Machining	Machining			
132	L-2		Machining			
133	I&P-2		Machining			
134	T-3		Machining			
135	UL-3	Machining	Machining			
136	L-3		Idle			
137	I&P-3		Machining			
138	T-1	Unloaded	Idle		Machining	
139	UL-1	Loaded				
140	L-1	Machining				
141	I&P-1	Unloaded	Idle		Machining	
142	T-2	Loaded				
143	UL-2	Machining	Machining			
144	L-2		Machining			
145	I&P-2		Machining			
146	T-3		Machining			
147	UL-3	Machining	Machining			
148	L-3		Idle			
149	I&P-3		Machining			
150	T-1	Unloaded	Idle		Machining	
151	UL-1	Loaded				
152	L-1	Machining				
153	I&P-1	Unloaded	Idle		Machining	
154	T-2	Loaded				
155	UL-2	Machining	Machining			
156	L-2		Machining			
157	I&P-2		Machining			
158	T-3		Machining			
159	UL-3	Machining	Machining			
160	L-3		Idle			
161	I&P-3		Machining			
162	T-1	Unloaded	Idle		Machining	
163	UL-1	Loaded				
164	L-1	Machining				
165	I&P-1	Unloaded	Idle		Machining	
166	T-2	Loaded				
167	UL-2	Machining	Machining			
168	L-2		Machining			
169	I&P-2		Machining			
170	T-3		Machining			
171	UL-3	Machining	Machining			
172	L-3		Idle			
173	I&P-3		Machining			
174	T-1	Unloaded	Idle		Machining	
175	UL-1	Loaded				
176	L-1	Machining				
177	I&P-1	Unloaded	Idle		Machining	
178	T-2	Loaded				
179	UL-2	Machining	Machining			
180	L-2		Machining			
181	I&P-2		Machining			
182	T-3		Machining			
183	UL-3	Machining	Machining			
184	L-3		Idle			
185	I&P-3		Machining			
186	T-1	Unloaded	Idle		Machining	
187	UL-1	Loaded				
188	L-1	Machining				
189	I&P-1	Unloaded	Idle		Machining	
190	T-2	Loaded				
191	UL-2	Machining	Machining			
192	L-2		Machining			
193	I&P-2		Machining			
194	T-3		Machining			

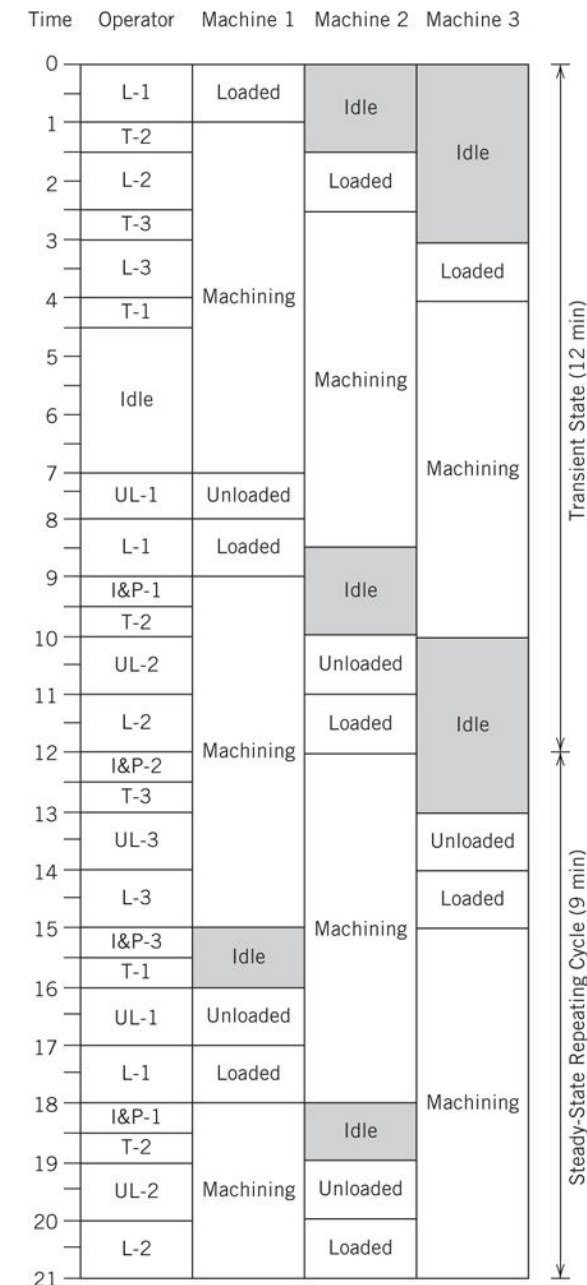
Machine Assignment Problems

Deterministic model (Example 2.7)

For the situation illustrated in Figure 2.18, it takes 0.5 minute to travel between machines, 1 minute to load a machine, 1 minute to unload a machine, 6 minutes of automatic machine time, and 0.5 minute to inspect and pack a finished part. As shown, the analysis begins with each machine empty and the operator standing in front of Machine 1 (M-1).

The operator loads M-1, walks to M-2, loads M-2, walks to M-3, loads M-3, walks to M-1, unloads M-1, loads M-1, inspects and packs the part removed from M-1, travels to M-2, and so forth.

As shown in Figure 2.18, it takes 12 minutes for the operator and three machines to achieve a steady-state condition; thereafter, a repeating cycle of 9 minutes in duration occurs. (In other words, if nothing interrupts the activities of the operator and the three machines, the 9-minute cycle will repeat indefinitely.)



Machine Assignment Problems

a = concurrent activity time (man and machine)

b = independent operator time

t = independent machine time

n' = ideal number of identical machines to assign an operator

m = number of identical machines assigned

T_c = repeating cycle time

I_o = idle operator time during a cycle

I_m = idle time for each machine during a cycle

Excluding idle time, each machine cycle requires $a + t$ minutes to complete a cycle. Likewise, the operator devotes $a + b$ minutes to each machine during a cycle. Hence, an ideal assignment is

$$n' = (a + t)/(a + b) \quad (2.6)$$

For the example, a equals two minutes, b equals one minute, and t equals six minutes. Therefore, n' equals 2.67 machines.

2 would cause the operator idle time

3 would cause the machine to have idle time

Machine Assignment Problems

Since a fractional number of machines cannot be assigned to an operator, consider what will happen if some integer number of machines, m , is assigned. The work content for the operator will total $m(a + b)$, while a machine cycle will be $(a + t)$ in duration. The repeating cycle will be the larger of the two, and the difference in the two will be idle time. If $m > n'$, then the repeating cycle will be $m(a + b)$ in duration; if $m < n'$, then the repeating cycle will be $(a + t)$ in duration. Therefore,

$$T_c = \begin{cases} (a + b) & m \leq n' \\ m(a + b) & m > n' \end{cases} \quad (2.7)$$

$$I_m = \begin{cases} 0 & m \leq n' \\ T_c - (a + t) & m > n' \end{cases} \quad (2.8)$$

$$I_o = \begin{cases} T_c - m(a + b) & m \leq n' \\ 0 & m > n' \end{cases} \quad (2.9)$$

For the example, with $m = 3$ and $n' = 2.67$, the repeating cycle is $3(2+1)$, or nine minutes as observed in Figure 2.18. The idle time for each machine during a repeating cycle equals $9 - (2+6)$, or one minute, again as observed in Figure 2.18. As shown in Figure 2.19, the work cell must be designed carefully to provide adequate access for incoming and outgoing material, machine maintenance, and operator movement.

You want to find out how many CNC Machines to assign to each operator. Time to unload one part and load the next part is 3 minutes. During this time the machine is stopped. It takes 20 minutes to run a part. The operator must inspect each part which takes 5 minutes. The inspection is done on the work bench. It takes 0.5 minutes to travel between machines.

Calculate the total cost for 2 and for 3 machines find the lower cost (min T_c):

For $m < n'$: $T_c(m) = (C_o + mC_m)(a+t)/m$

For $m > n'$: $T_c(m) = (C_o + mC_m)(a+b)$

The cost to run the machine is \$55/hr

The cost of an employee is \$40/hr

2 machines per employee:

$m=2$

$TC = (C_o + mC_m)(a+t)/m = (40+255)*(3+20) / 2. = 1725$

• min at $m=2$

3 machines per employee

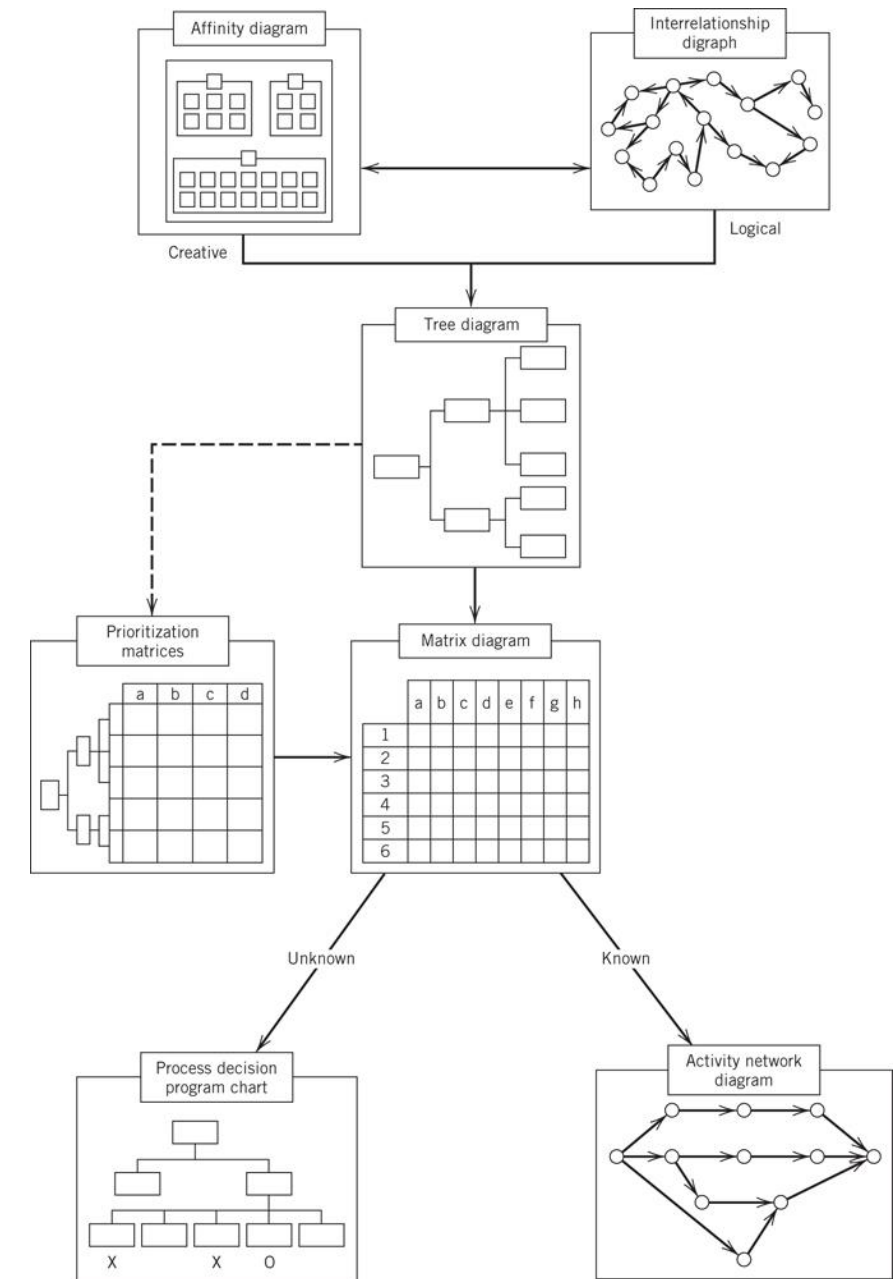
$m = 3$

$TC = (C_o + mC_m)(a+b) = (40+355)(3+5.5) = 1745.5$

The **Seven** management and planning tools

Gained acceptance as a methodology for improving planning and implementation:

- Affinity diagram
- Interrelationship digraph
- Tree diagram
- Matrix diagram
- Contingency diagram
- Activity network diagram
- Prioritization matrix



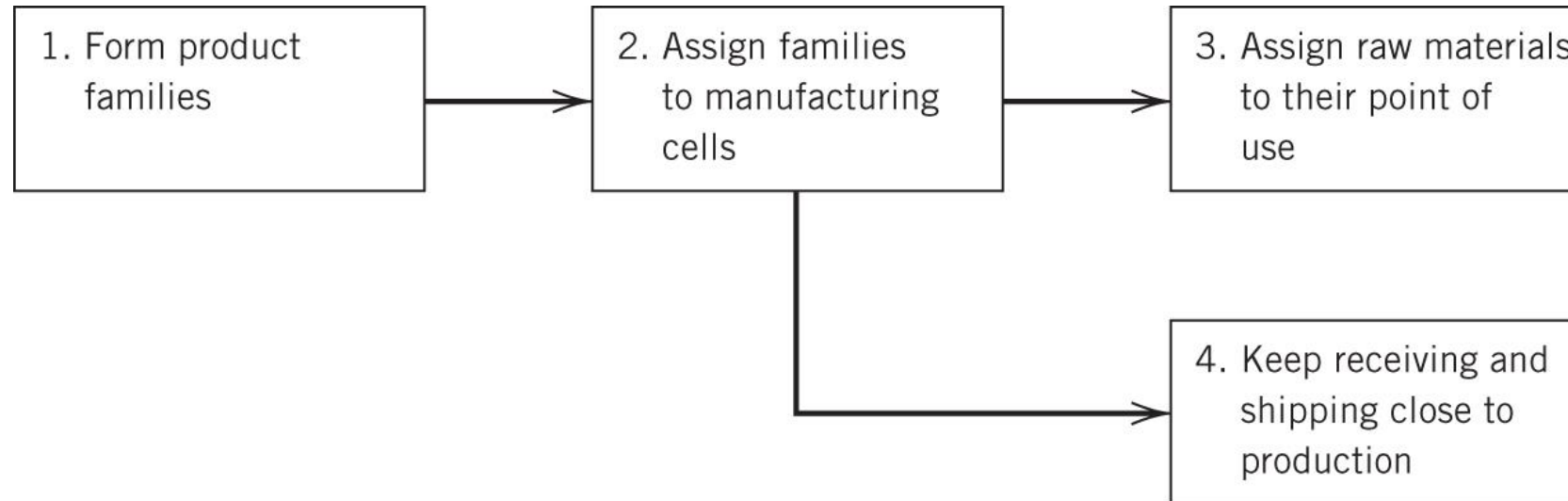
Affinity Diagram

- Verbal data for the system/process under study.
- Take discussion to reduce manufacturing lead time

Issues in reducing manufacturing lead time				
Facilities design	Equipment issues	Quality	Setup time	Scheduling
1. Form product families	1. Operator certification program	1. Provide training on how to use process documentation	1. Provide documentation on setup procedures	1. Provide visibility to daily product sequence
2. Assign families to cells	2. Sit technicians closer to production	2. Implement successive inspection with feedback	2. Locate fixtures and tooling close to machines	2. Do not authorize products for which the needed parts are not available
3. Assign raw mtl's to their point of use	3. Monitor breakdowns to predict future occurrences	3. Develop mistake-proof devices	3. Provide training so operators can participate	3. Negotiate frequent and smaller lots to customers
4. Keep receiving and shipping close to production	4. Recruit enough technicians per shift	4. Develop capabilities for monitoring key machine parameters	4. Provide information on daily sequence	

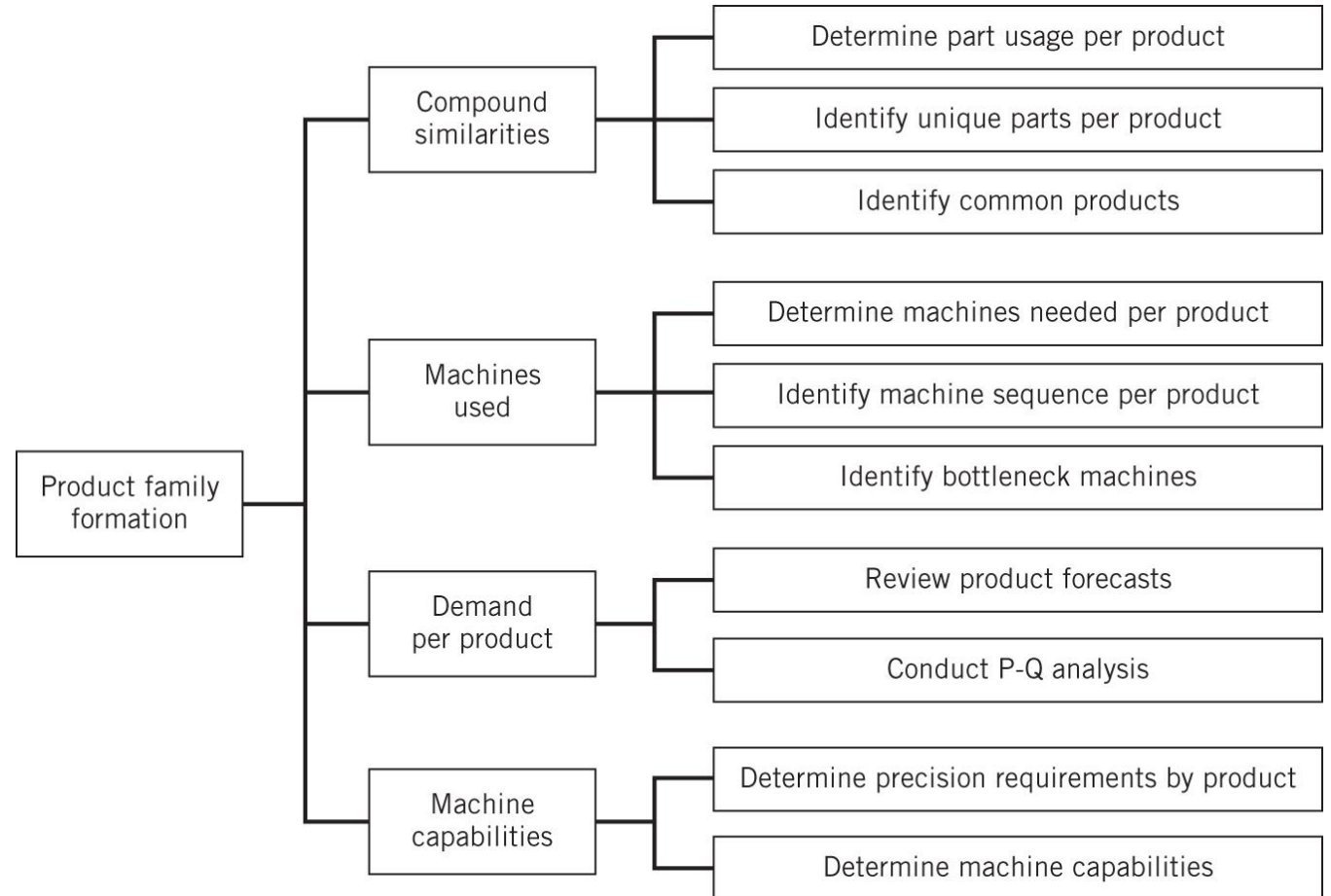
Interrelationship Digraph

- Map the logical links among related items, and identify which items impact others the most.
- Take the facilities design tab in the affinity diagram for example.



Tree Diagram

- Map in increasing detail the actions to be performed for an objective.
- For instance, to form the product family in the interrelationship diagram:



Matrix Diagram

- Organize information into sets of characteristics, functions, and tasks for comparison.
- E.g.: the table below summarizes the participants and their role within the small teams (to allocate the jobs).

Team\Participants	Joe	Mary	Jerry	Lou	Linda	Daisy	Jack
Part usage team	P	C	P	L			P
Machine use & cap team	L		C				P
Demand forecast team				P	C	L	

Note:

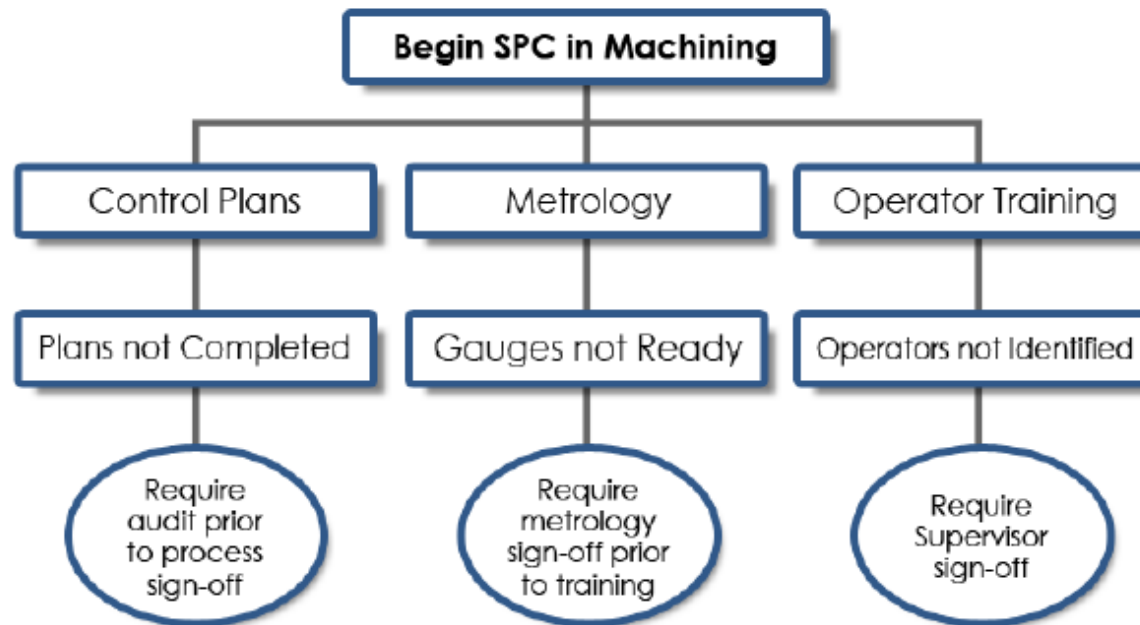
L: Team Leader

C: Team Coordinator

P: Team Participant

Contingency Diagram (Process Decision Program Chart)

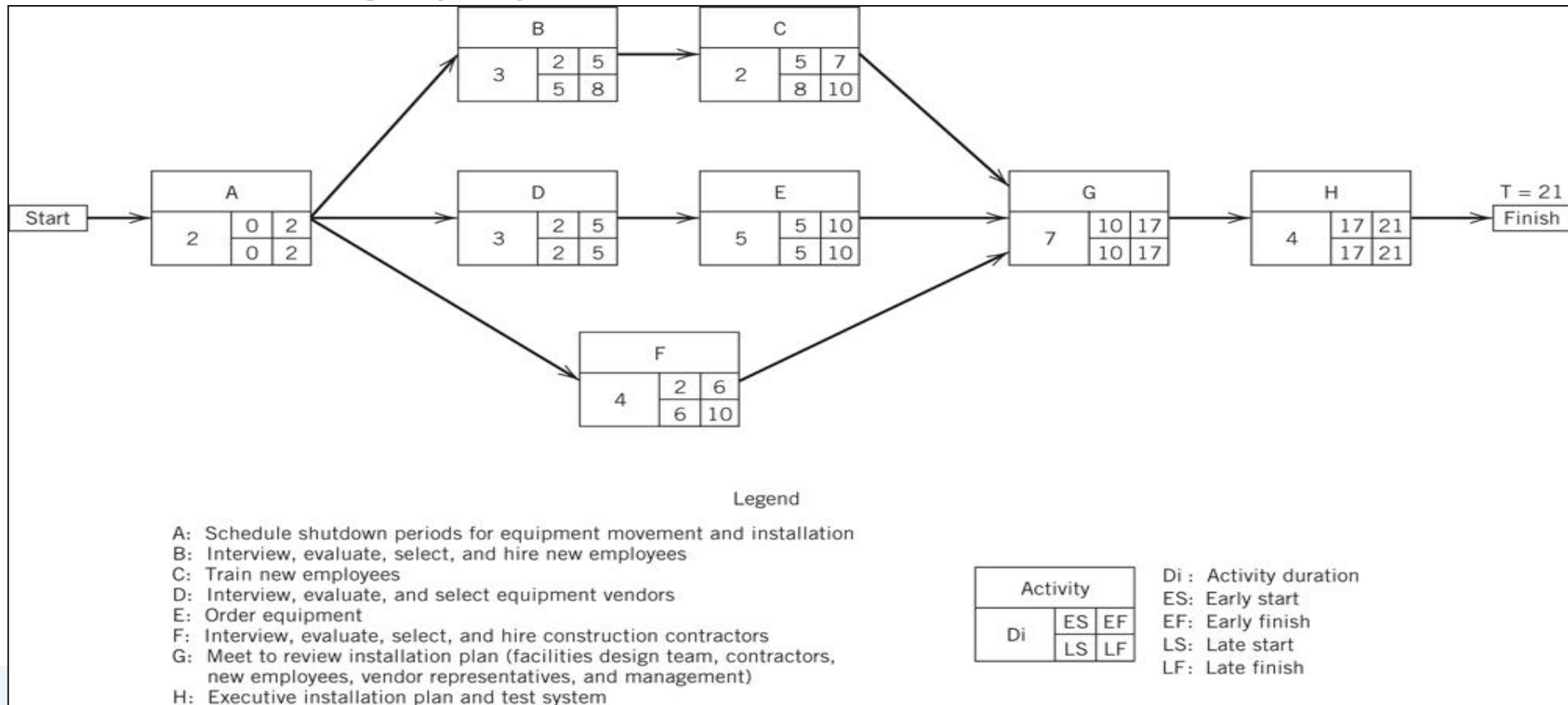
- Map conceivable events and contingencies that might occur during implementation
- Useful when the project being planned includes unfamiliar tasks



<http://www.sixsigmadaily.com/process-decision-program-charts-pdpc/>

Activity Network Diagram

- Develop a *work schedule* for the facilities design effort (can be replaced by Gantt chart).
- A well-thought timetable is needed to understand the length of the facilities design project.



Prioritization Matrix

- Be used to judge the relative importance of each criterion as compared to the others

		Criteria being compared to								Row	
		Low cost	Use of technology	Potential Savings	Increased speed	Decreased defects	Customer Satisfaction	Minimal impact	Easy to implement	Quick Results	
Criteria		a.	b.	c.	d.	e.	f.	g.	h.	i.	
a.	Low Cost		5								
b.	Use of Technology	0.2									
c.	Potential Saving										
d.	Increased Speed										
e.	Decreased Defects										
f.	Customer Satisfaction										
g.	Minimum Impact										
h.	Easy to Implement										
i.	Quick Results										
Column Total											

Prioritization Matrix

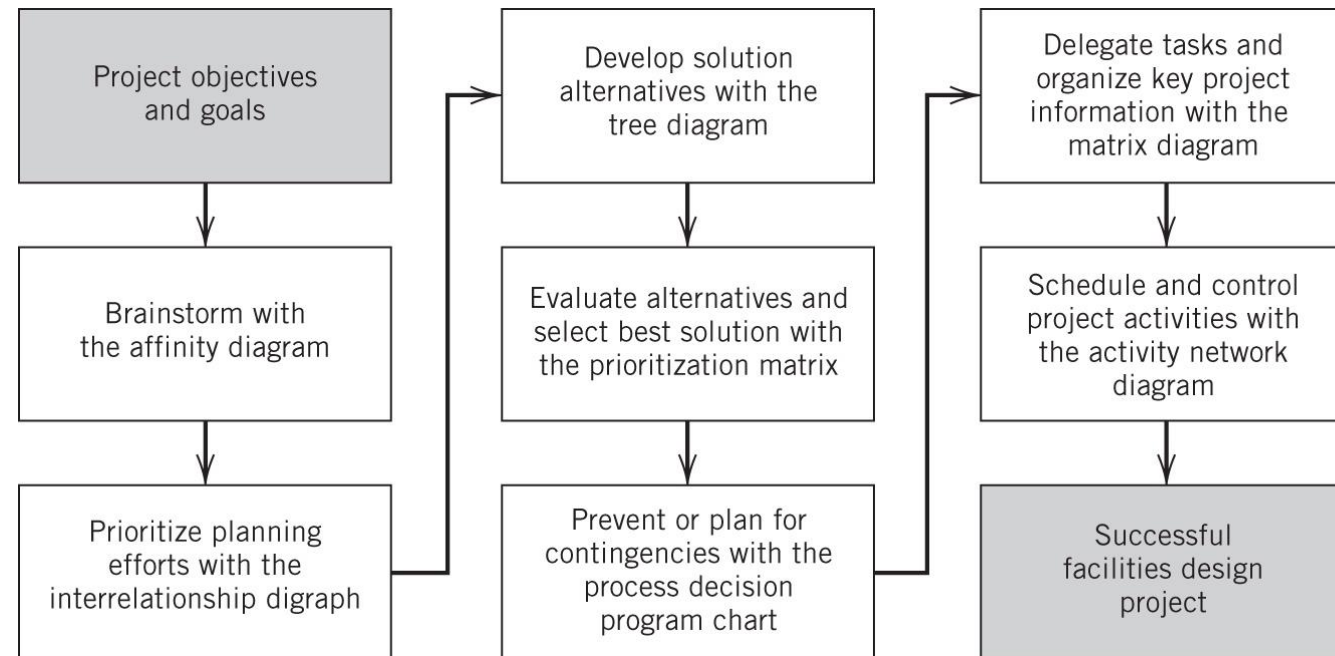
- Example:

Criteria	Cost effective	Decreased defects	Increased productivity	User friendly		
↓ Options / Weight →	0.85	0.8	1.75	0.6	Weighted scores	Rank
New equipment 1	3	2	2	3	9.45	3
New equipment 2	2	3	3	4	11.75	1
New equipment 3	1	2	4	2	10.65	2
New equipment 4	4	1	1	1	6.55	4

<https://citoolkit.com/articles/prioritization-matrix/>

Seven Management and Planning Tools

- All tools can be used to support an integrated facilities design process, or they can also be used independently.
- An example of the applications of these tools:





Introduction

Three important considerations in determining facility requirements

- Flow systems
 - Movements of goods, materials, energy, information, and/or people
- Activity relationships
 - Quantitative or qualitative measure of the relationships between departments
- Space requirements
 - Determination of space requirements for facility, department, workstations, etc.

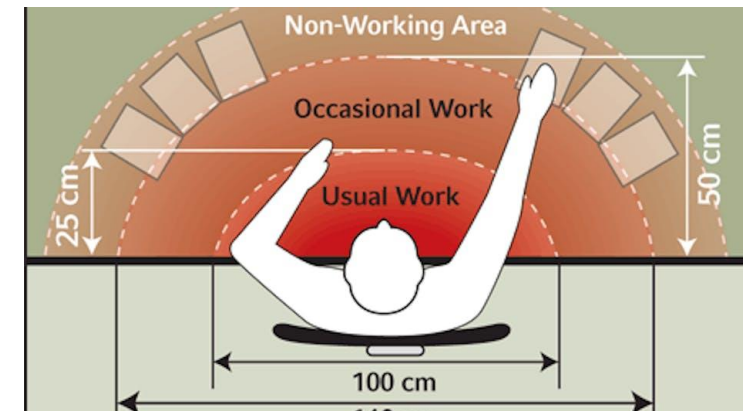
Flow Systems

- Flow is the movement of goods, materials, energy, information, people
- Product Flow **P**rocess
 - Movement of refrigerators from manufacturer through distribution centers, and ultimately to customer
- Information Flow **P**rocess
 - Transmission of sales orders from sales department to production control department
- Flow Process Involving **P**eople
 - Movement of patients, staff, visitors at hospital

Types of Flow Processes

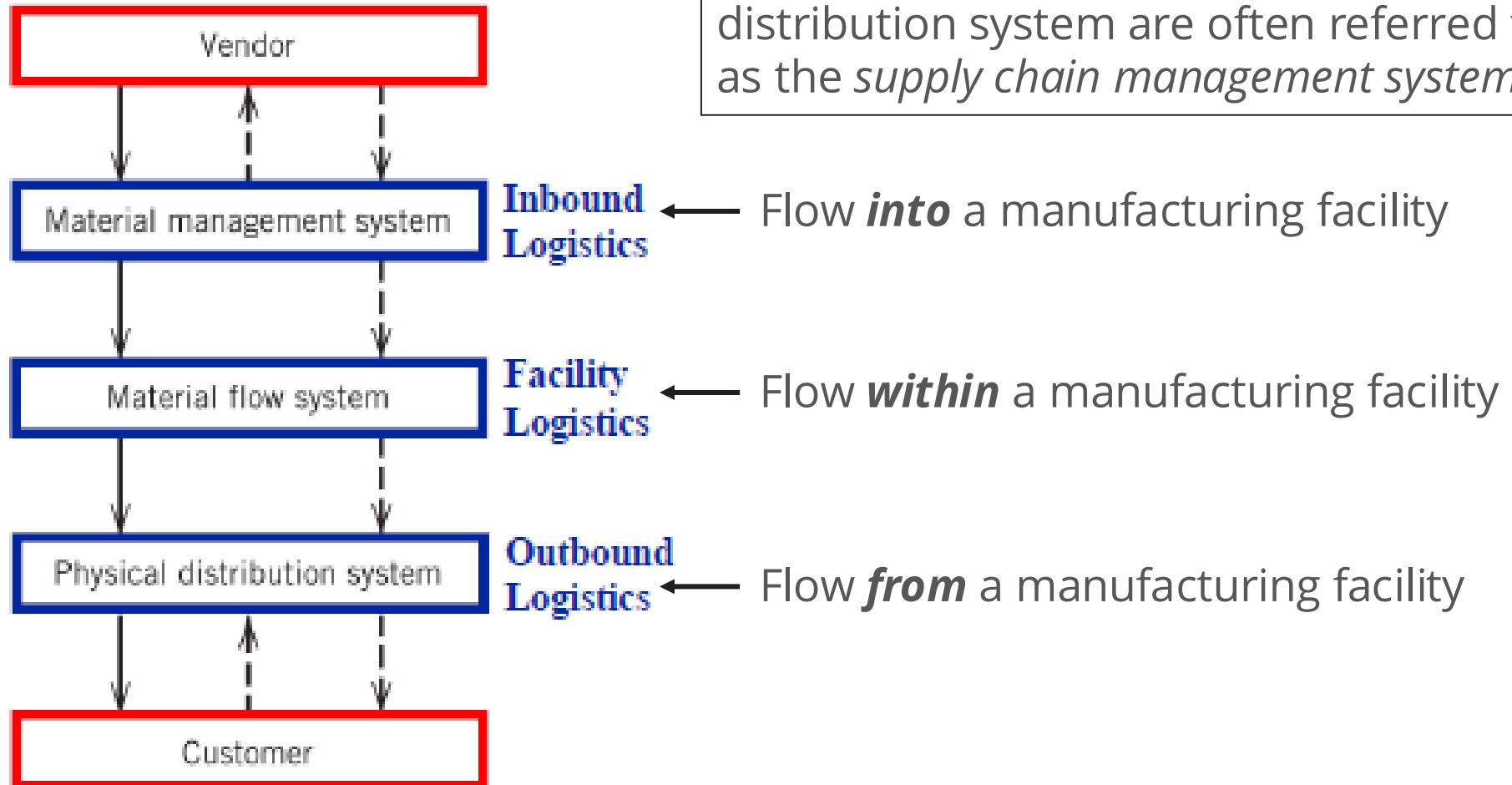
- Discrete flow processes (primary emphasis in this course): discrete items move through the flow process, such as machine, products, cans, etc.
- Continuous flow processes: products continuously move through successive production states, such as electric current, oil, etc.
- Discrete flow systems can be categorized supply, manufacture, and distribution stages:
 - Materials management system
 - Material flow system
 - Physical distribution system

Logistics system



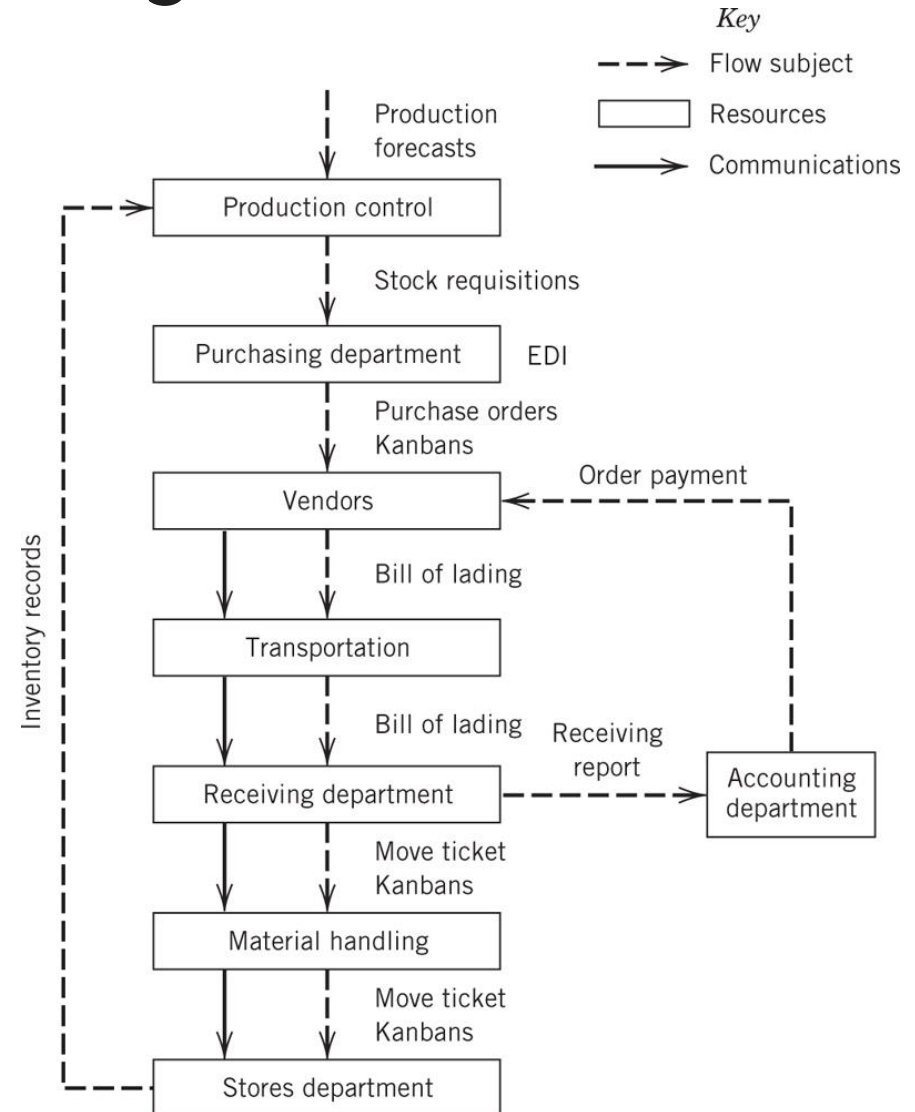
Flow and Data

- Logistics system



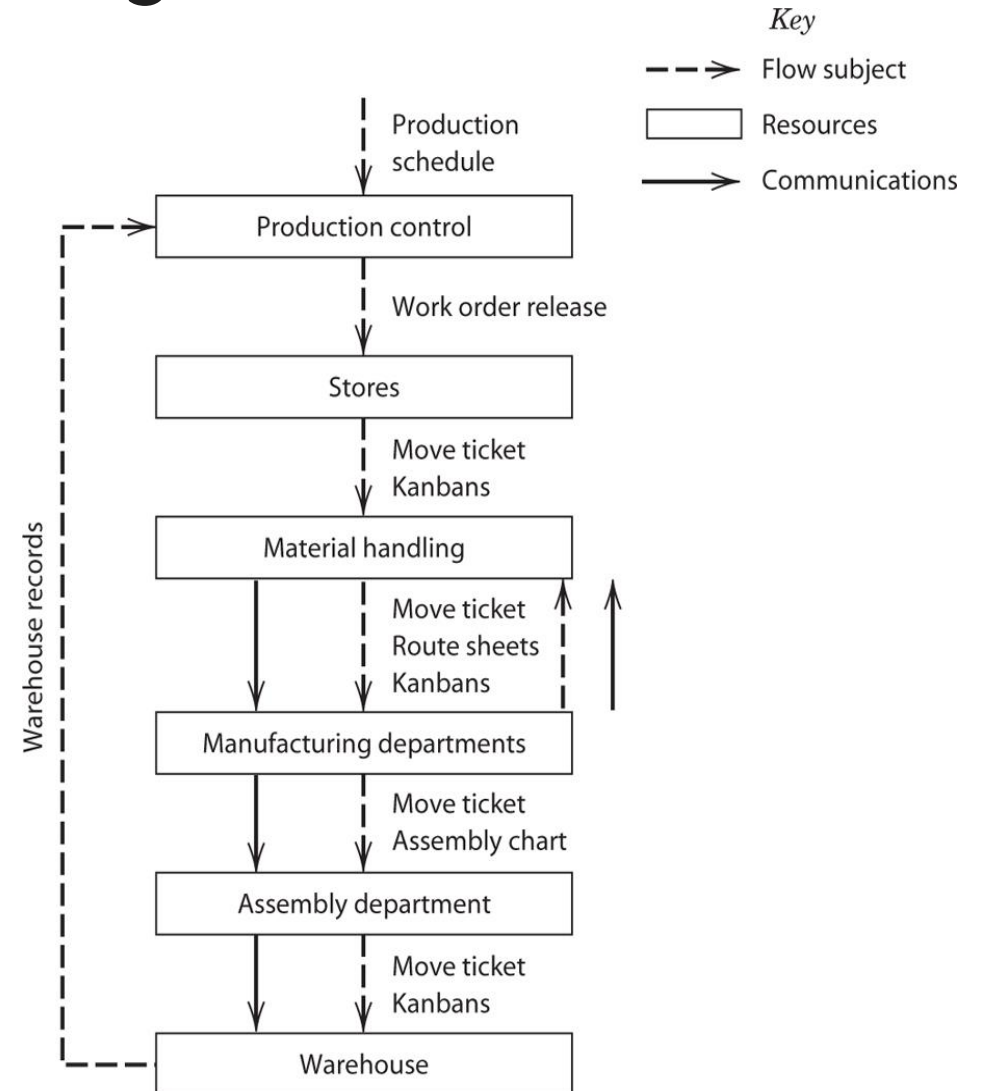
Materials Management System

- Flow of materials **into** a manufacturing facility
- Resources: production control and purchasing functions; vendors; transportation and material handling equipment; receiving, storage, and accounting functions.
- Communications: production forecasts, inventory records, stock requisitions, purchase orders, bills of lading, electronic data interchange (EDI), etc.



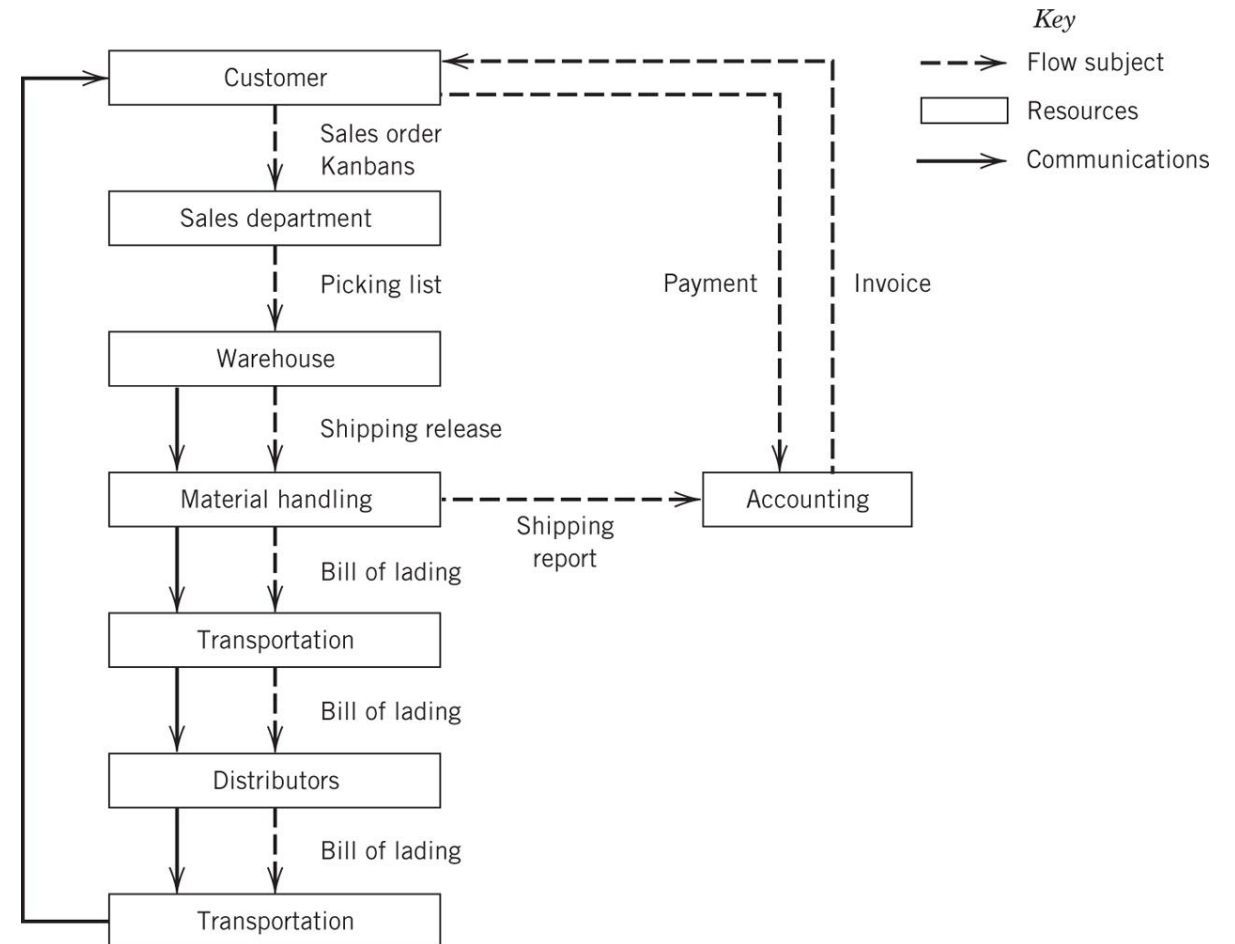
Materials Management System

- Flow of materials, parts, and supplies **within** a manufacturing facility.
- Resources: production control and quality control departments; manufacturing, assembly, and storage departments; material handling equipment; factory warehouse.
- Communications: production schedules, work order releases, move tickets, kanbans, etc.



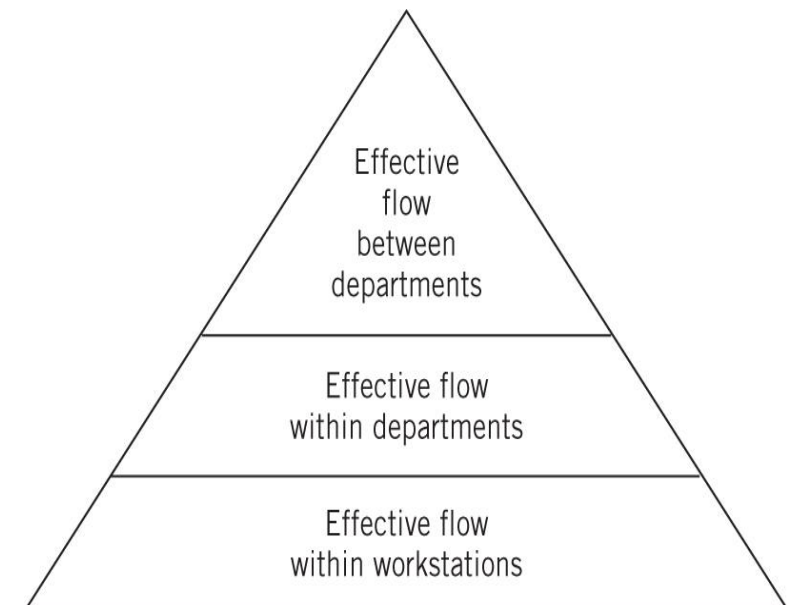
Physical Distribution System

- Flow of products **from** a manufacturing facility, the subjects are the finished goods produced by a firm.
- Resources: customer; sales and accounting departments, and warehouses; material handling and transportation equipment; distributors
- Communications: sales orders, packing lists, shipping reports, etc.



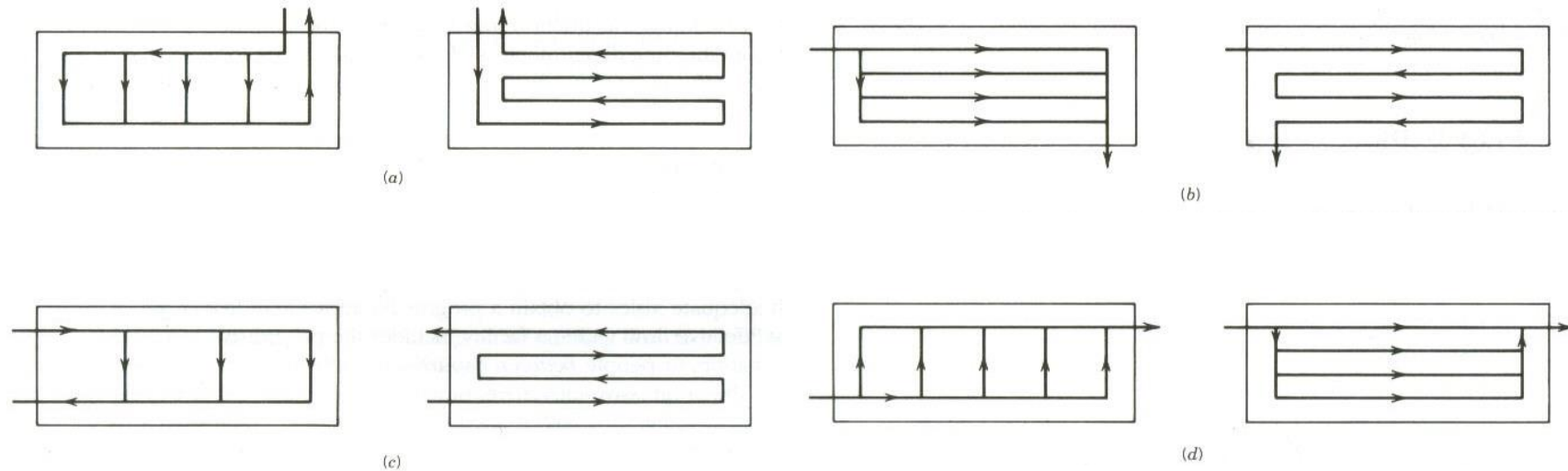
Flow Planning

- Process of arranging activities in combinations of basic flow patterns (based on flow analysis), both quantitative (from-to chart) and qualitative (activity relationship diagram).
- Hierarchical decomposition (material flow system)
 1. Flow between departments (inter-cell)
 2. Flow within product departments (intra-cell)
 3. Flow within workstations



Flow Planning

- Flow between departments (inter-cell)
 - Is used to evaluate overall flow within a facility
 - Flow patterns determines the entrance (receiving) and exit (shipping)



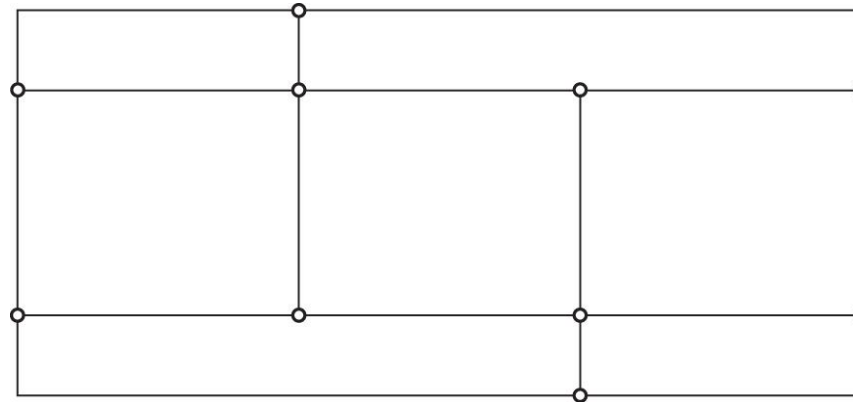
Flow within a department considering the locations of input/output points.
(a) At the same location. (b) On adjacent sides. (c) On the same side. (d) On opposite sides

Flow Planning

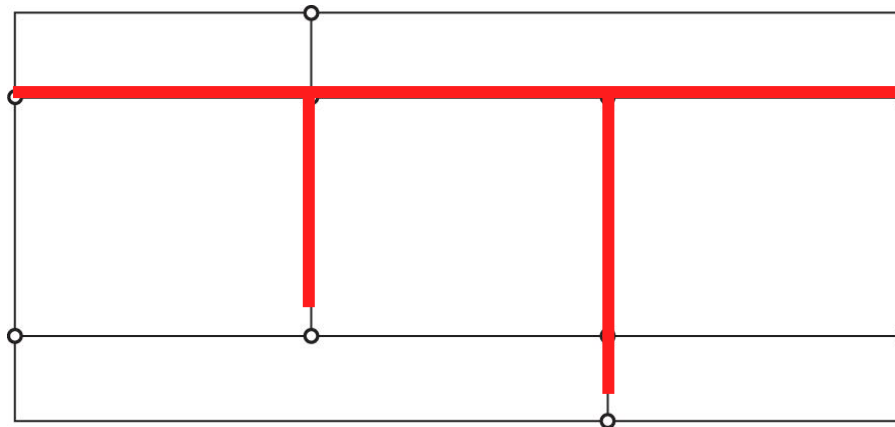


Flow Planning

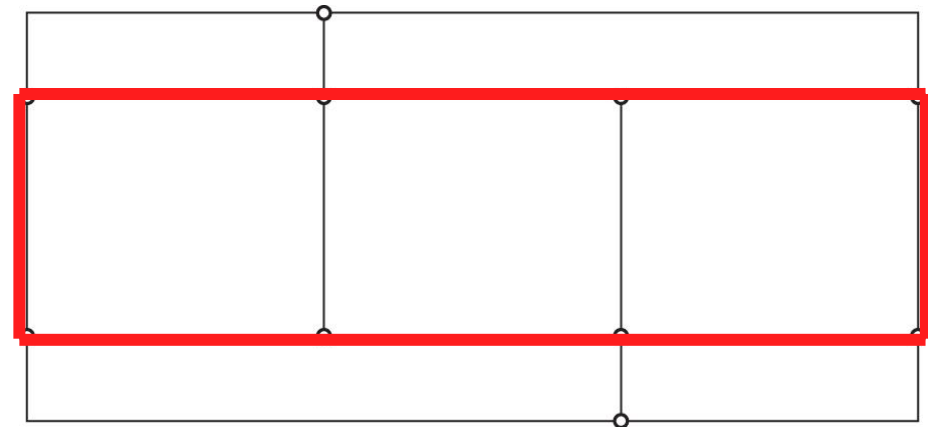
- Determination of best structure requires mathematical models and simulation – will be introduced later.



Conventional between-department flow structure



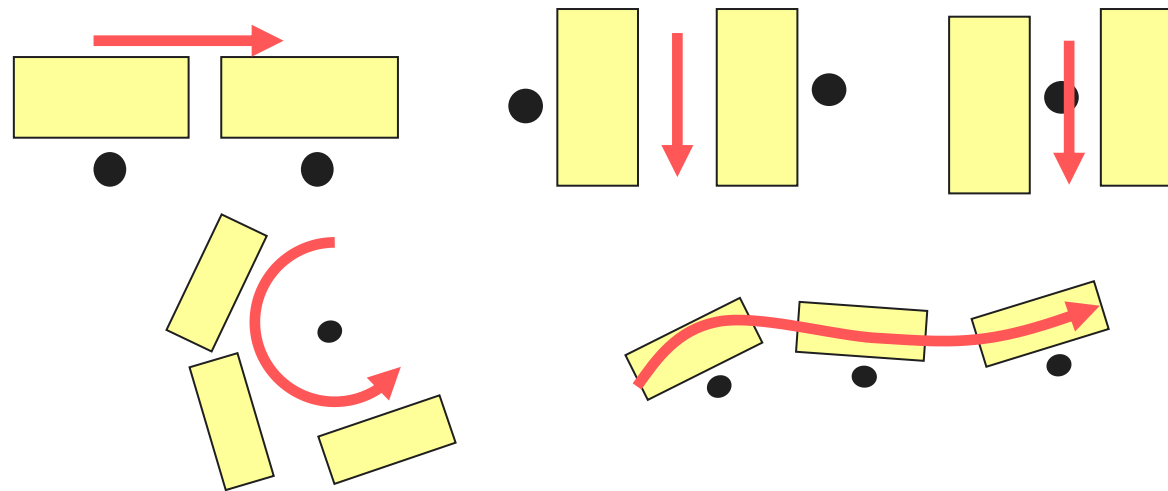
Spine between-department flow structure



Loop between-department flow structure

Flow Planning

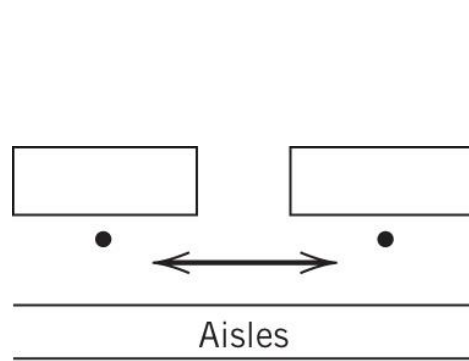
- Flow within a department (intra-cell)
- Depends on the type of department (product, process, product family, etc.)
- Flow patterns



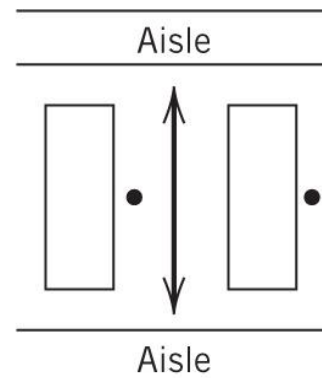
Flow within product departments. (a) End-to-end. (b) Back-to-back. (c) Front-to-front. (d) Circular. (e) Odd-angle. Dots: operator.

Flow Planning

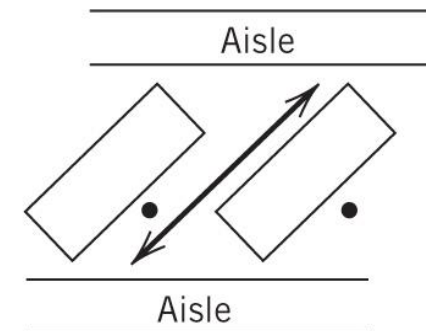
- Flow within a process department
 - Similar or identical machines are grouped in the same department, where minimal amount of flow should occur
 - Workstation – aisle arrangements
 - Depends upon available space, interactions among workstations, and size of materials to be handled



1 machine/operator
END-TO-END



1 machine/operator
BACK-TO-BACK

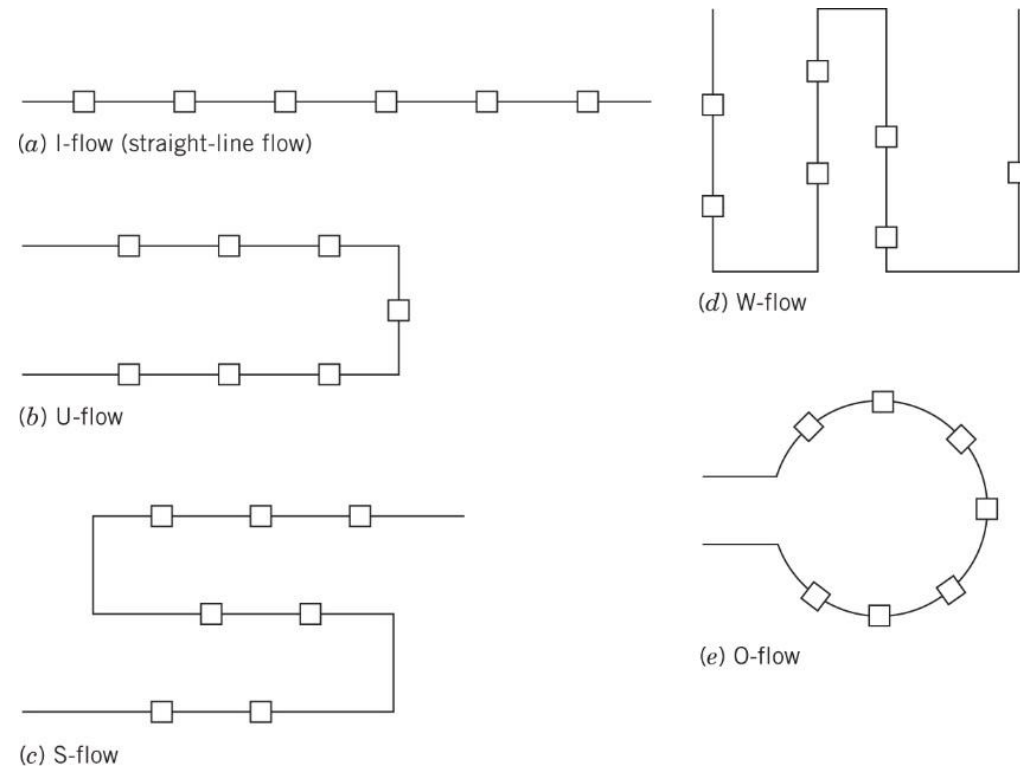


2 machine/operator
FRONT-TO-FRONT

Flow within process departments. (a) Parallel. (b) Perpendicular. (c) Diagonal.

Flow Planning

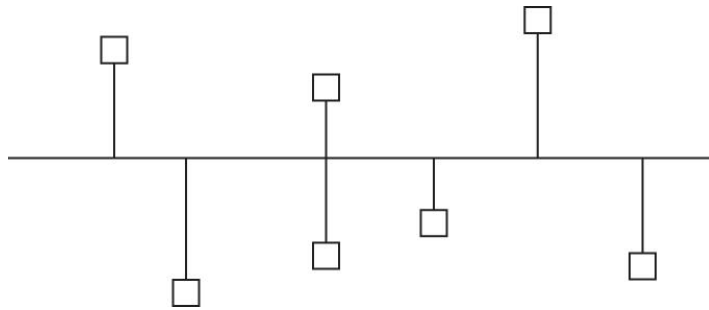
- Flow within product and process departments considering material handling.



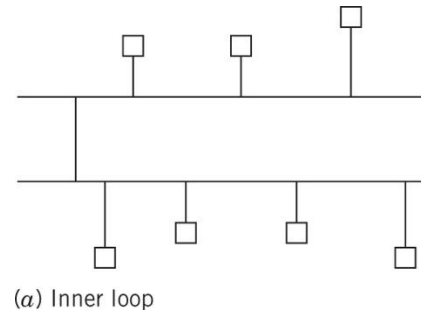
The line flow patterns

Flow Planning

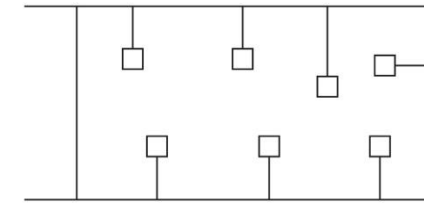
- Other flow patterns considering material handling:



The spline flow pattern

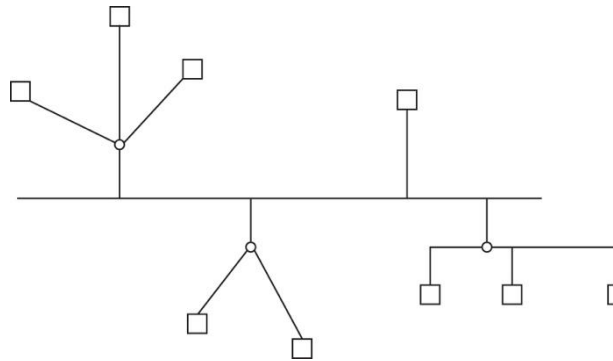


(a) Inner loop



(b) Outer loop

The loop flow pattern



The tree flow pattern

Flow Planning

- Flow within workstations
 - Motion studies and ergonomic considerations are important in establishing the flow within workstations.
 - Flow should be simultaneous, symmetrical, natural and rhythmical.



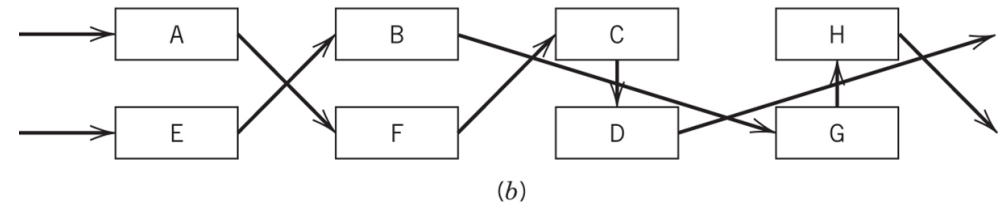
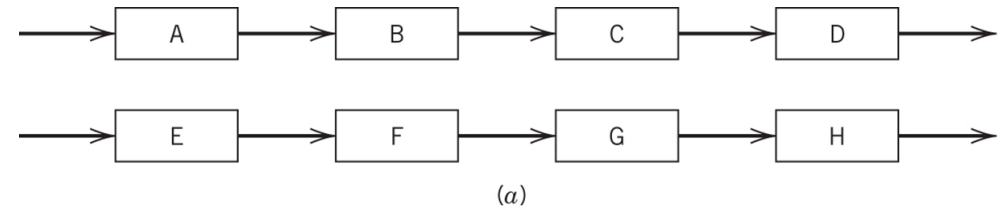
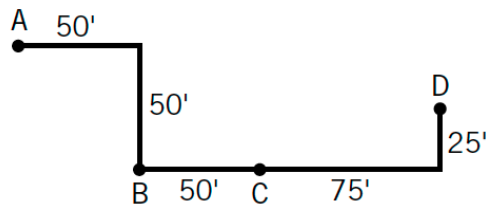
Motion Study

Flow Planning

- Principle 1:
 - Uninterrupted (no intersection with other paths) flow path directly from origin to destination
 - No backtracking

- Example: backtracking impacts

Compare A – B – C – D vs. A – B – A – C – D



Flow path A – B – C – D
 $(50' + 50') + 50' + (75' + 25') = 250 \text{ feet}$

Flow path A – B – A – C – D
 $(50' + 50') + (50' + 50') + (50' + 50') + 50' + (75' + 25') = 450 \text{ feet}$

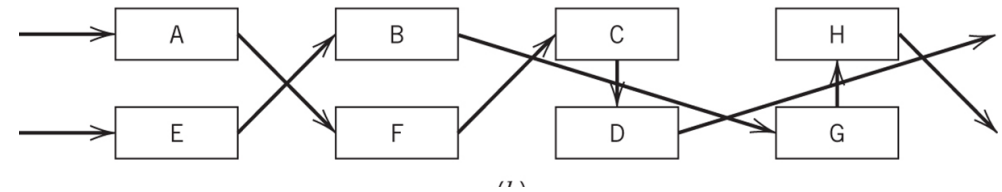
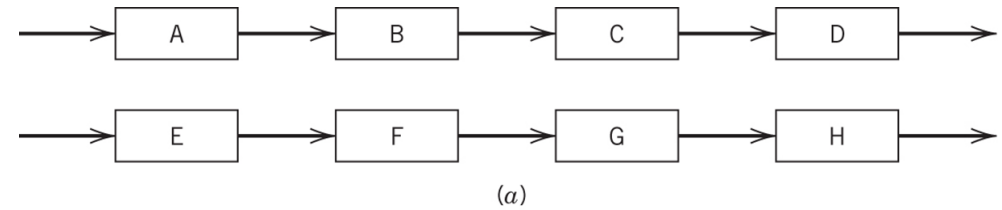
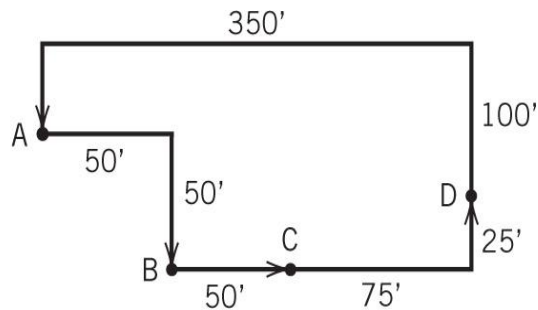
Backtracking penalty

Flow Planning

- Principle 1:
 - Uninterrupted (no intersection with other paths) flow path directly from origin to destination
 - No backtracking

- Example: backtracking impacts

Compare A – B – C – D vs. A – B – A – C – D



For path A – B – C – D

$$\text{Total path} = (50' + 50') + (50') + (75' + 25') = 250 \text{ feet}$$

For path A – B – A – C – D

$$\text{Total path} = (50' + 50') + [50' + (75' + 25') + 450'] + (50' + 50') + 50' + (75' + 25') = 950 \text{ feet}$$

Backtracking penalty

Flow Planning

- Principle 2:
 - Deliver directly to the point of use - eliminate waste
 - Plan appropriate unit of load, use pallets to minimize trips
 - Combine flows and operations, e.g. automobile assembly
- Principle 3:
 - Minimize travel distances
 - Mechanize or automate transfer

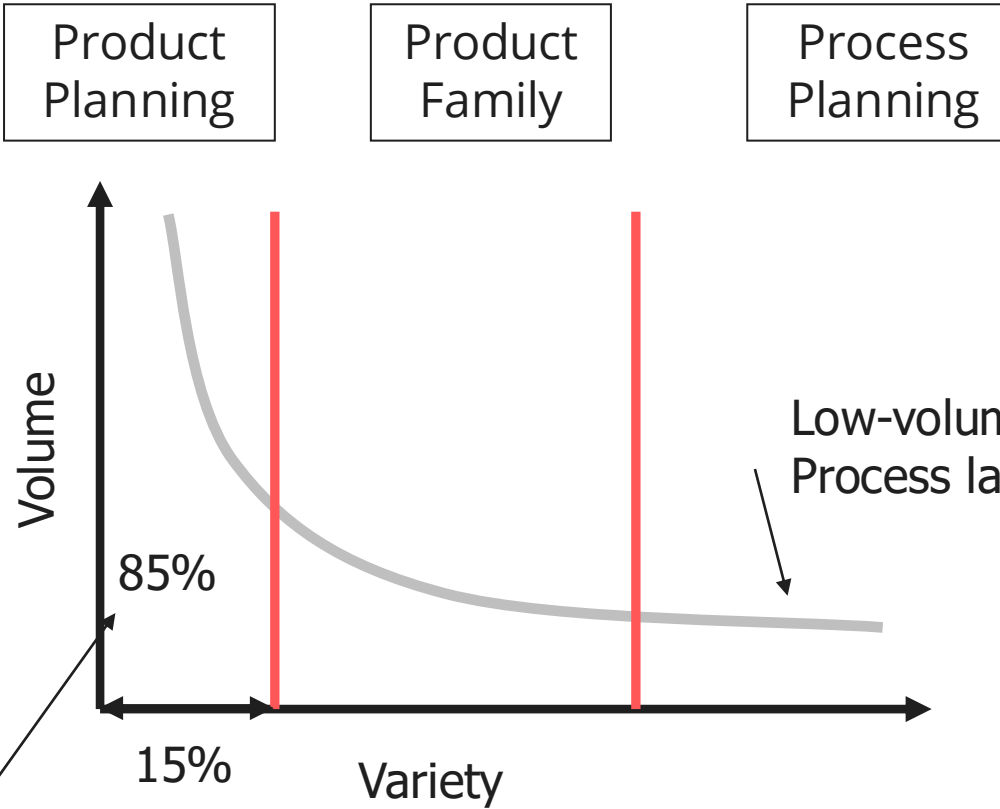
Achieving Effective Flow

- Must consider flow patterns and structures
- Adequate aisles to accommodate desired flow
- Must consider flow
 - Between departments
 - Within departments
 - Within workstations

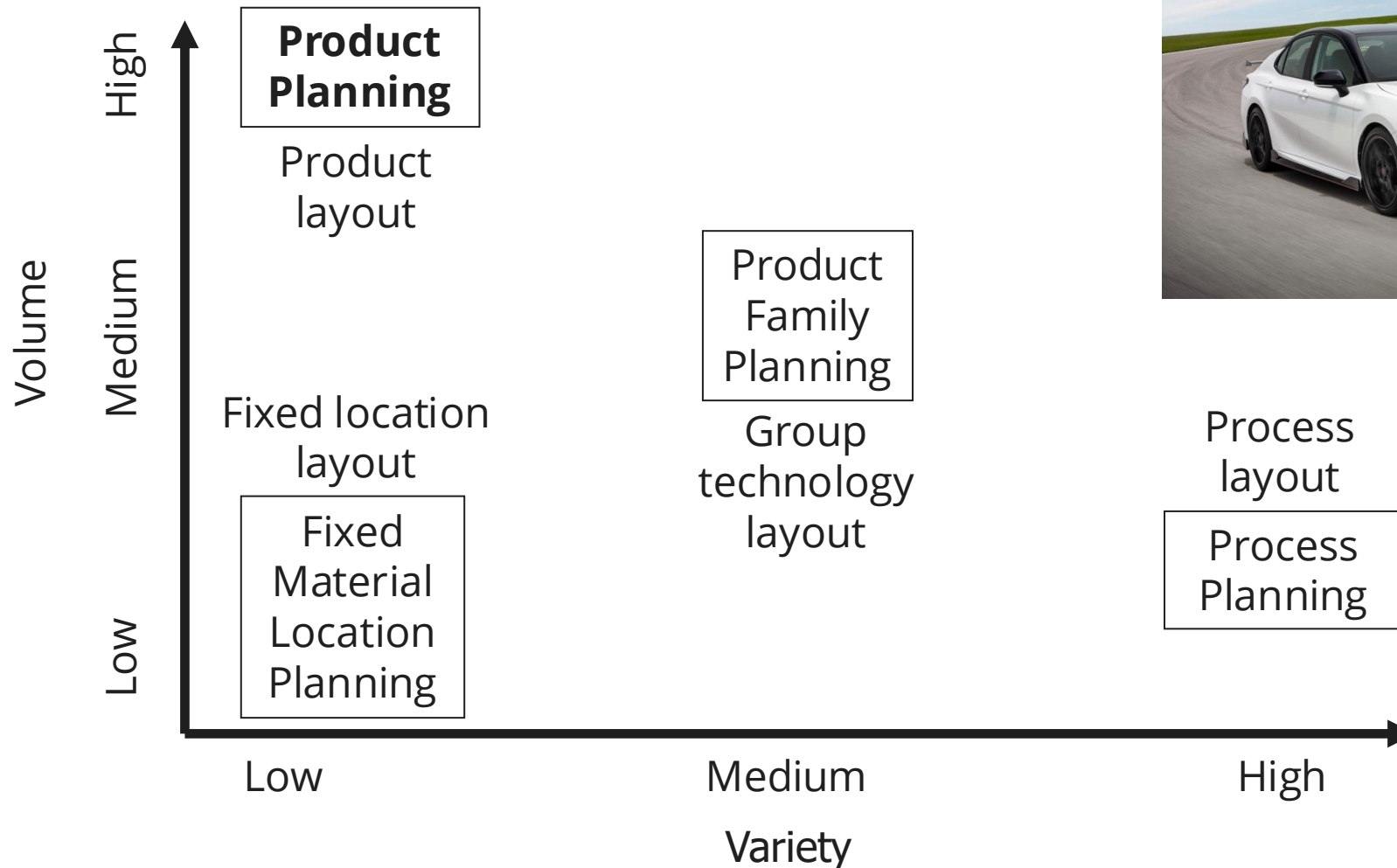
Departmental Planning

- **Fixed material location:** product is fixed, materials/machines are moved. For very large products.
 - Examples: Shipbuilding, aerospace
- **Product planning:** product line departments, flow shop. For standardized products with large & stable demand.
 - Examples: Automotive industry
- **Product family (Group layout):** group technology/cellular manufacturing. Parts can be divided into part families.
 - Examples: gears, nut/screws
- **Process:** layout of resources by function, job-shop. Small to medium-sized metal cutting industry.
 - Examples: job shop

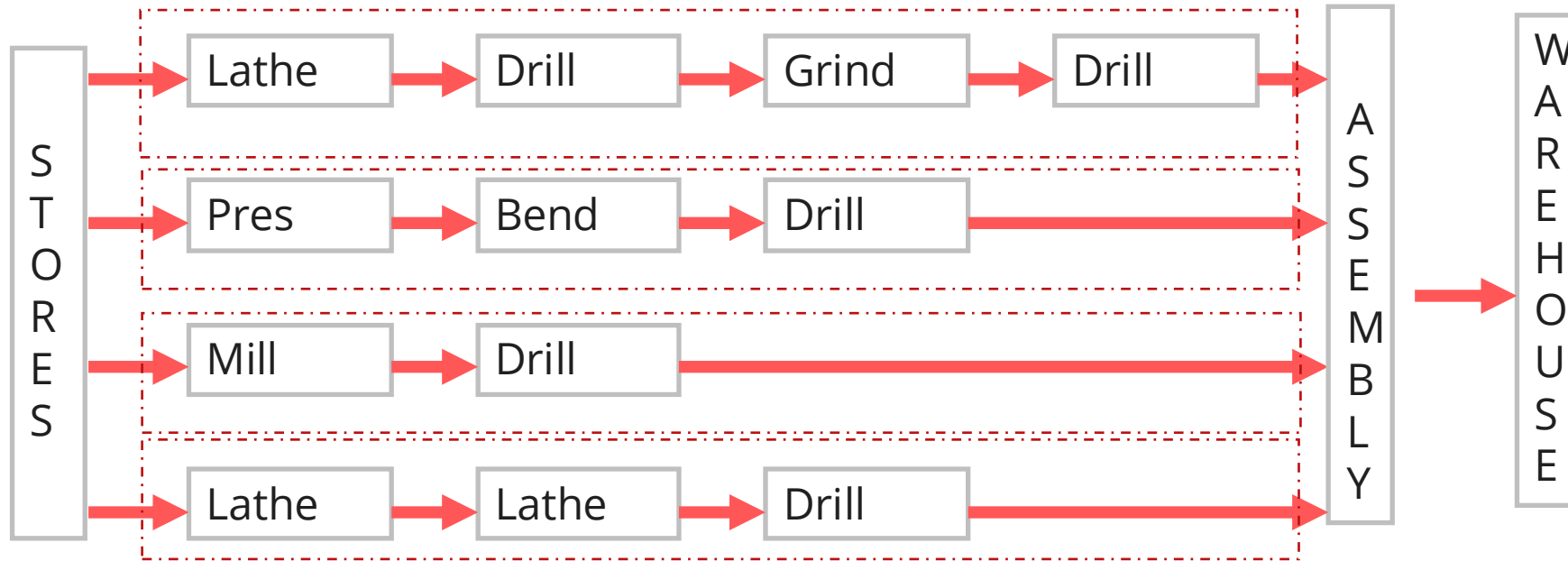
Departmental Planning



Departmental Planning



Product Layout



(a) Product layout

- Based on processing sequences online
- Automobile
- Materials flow from workstation to next
- High volume, low variety
- “flow shop” layout

We have one product department for Product A, but it has 4 operations or stations (since drill is repeated, we may also say the department for Product A has 3 operations or stations)

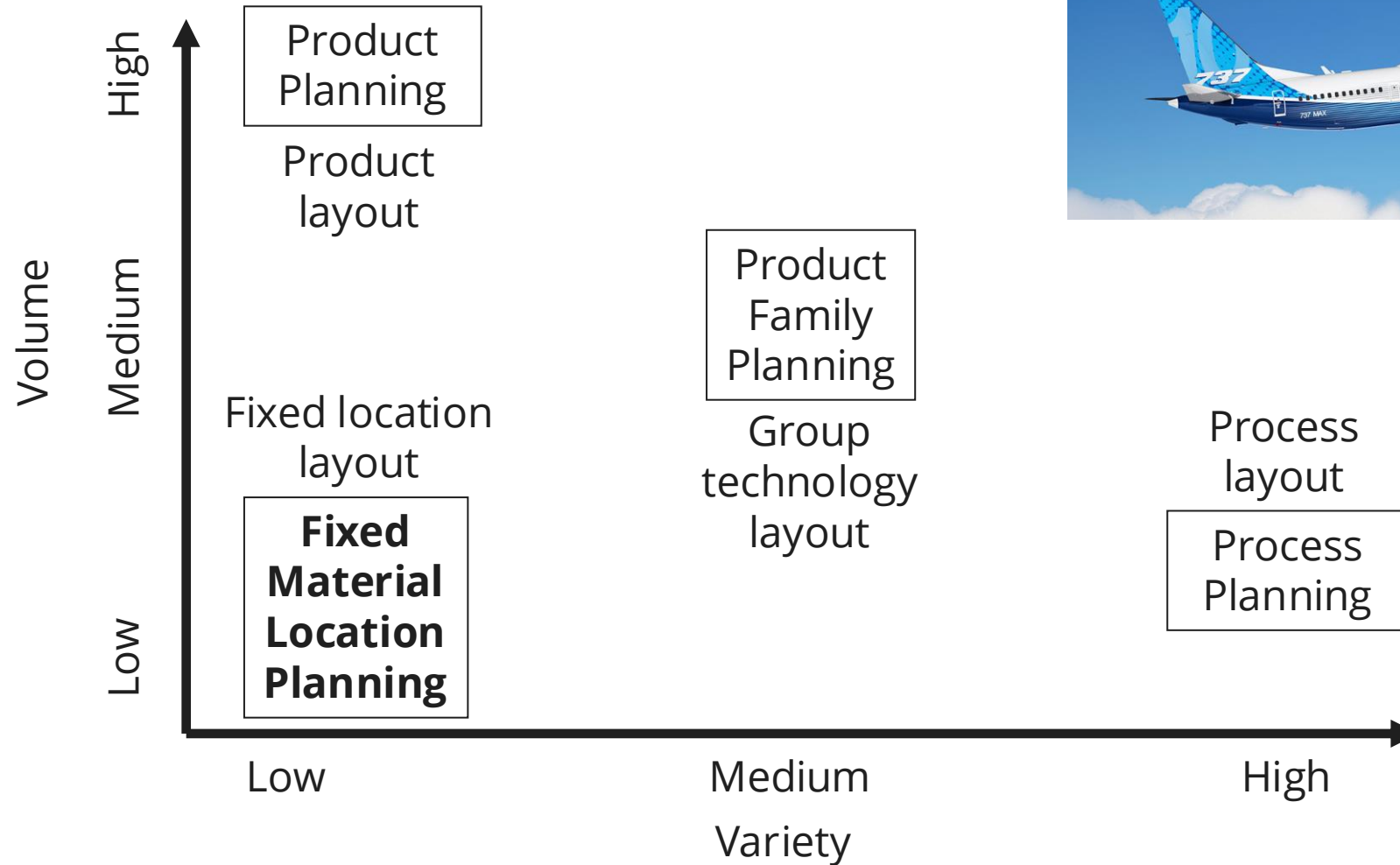
Advantages and Limitations of Layouts

Product layout

Advantages	Limitations
1. Smooth, simple, logical, and direct flow lines result. 2. Small work-in-process inventories should result. 3. Total production time per unit is short. 4. Material handling requirements are reduced. 5. Less skill is required for personnel. 6. Simple production control is possible. 7. Special-purpose equipment can be used.	1. Machine stoppage stops the line. 2. Product design changes cause the layout to become obsolete. 3. Slowest station paces the line. 4. General supervision is required. 5. Higher equipment investment usually results.

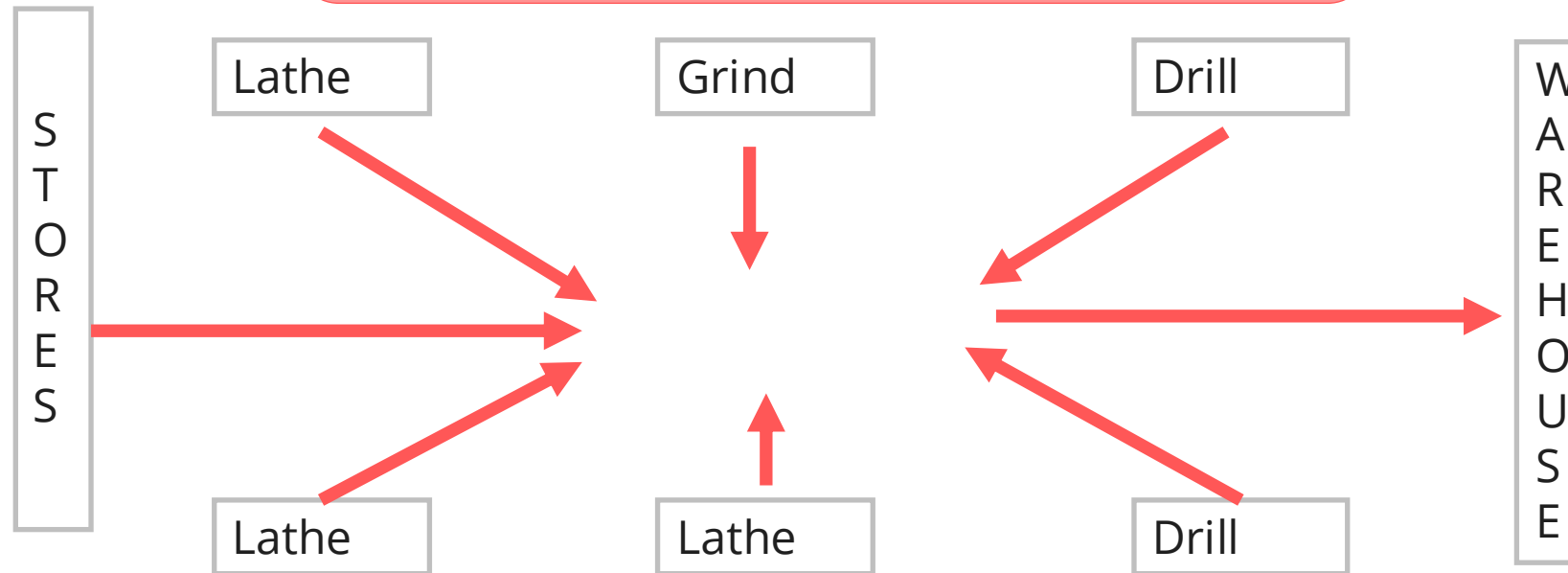
Source: Facilities Planning (Tompkins, White, Bozer, Tanchoco) Table 3.2, pp.112-113

Departmental Planning



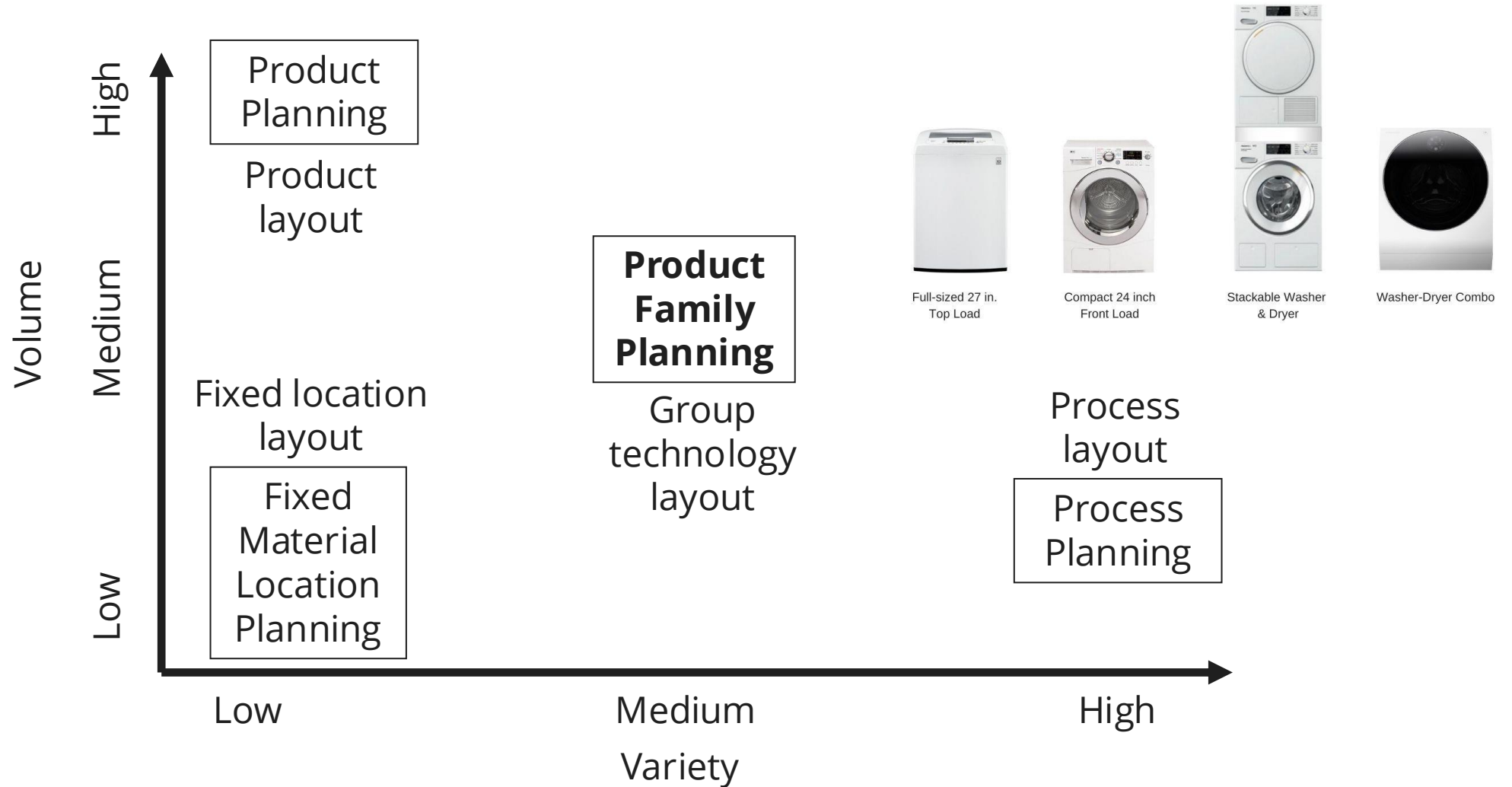
Fixed Location Layout

- Workstations are brought to material
- Aircraft assembly, shipbuilding, construction
- Sequencing, placement of workstations
- Low volume, low variety

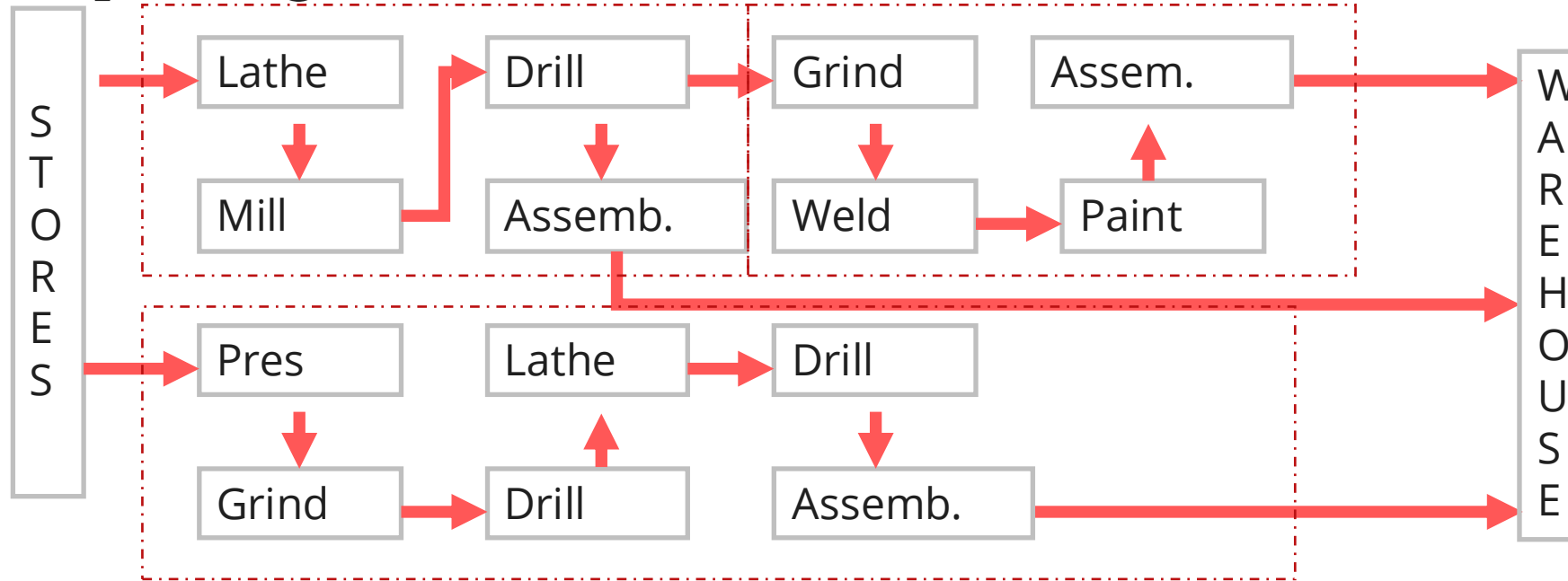


(b) Fixed location layout

Departmental Planning



Group Layout



(c) Group layout

- Grouping of non-identical parts to form production families (pseudo-product)
- Common processing sequences
- Medium variety, medium volume
- Electronics
- “batch” production

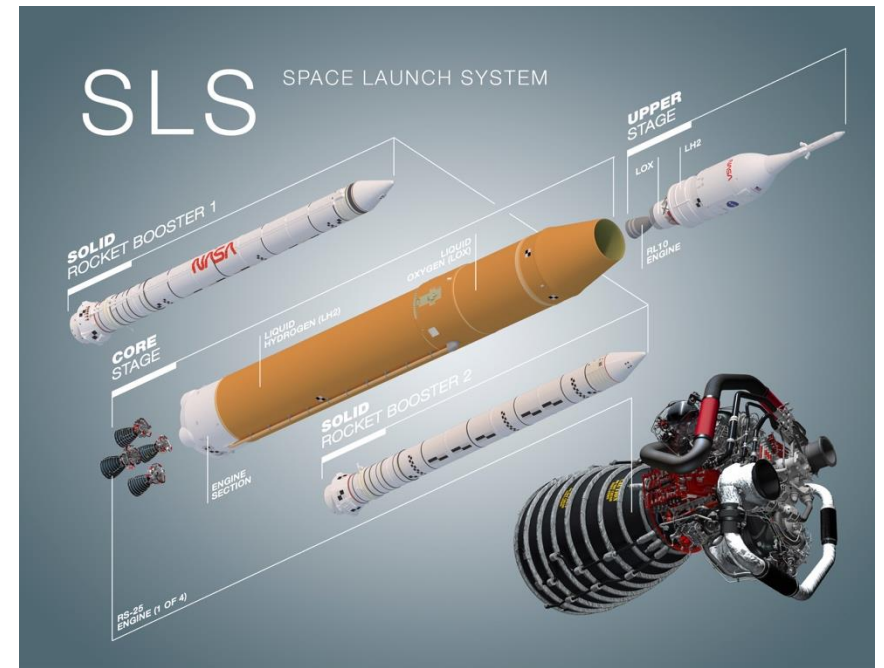
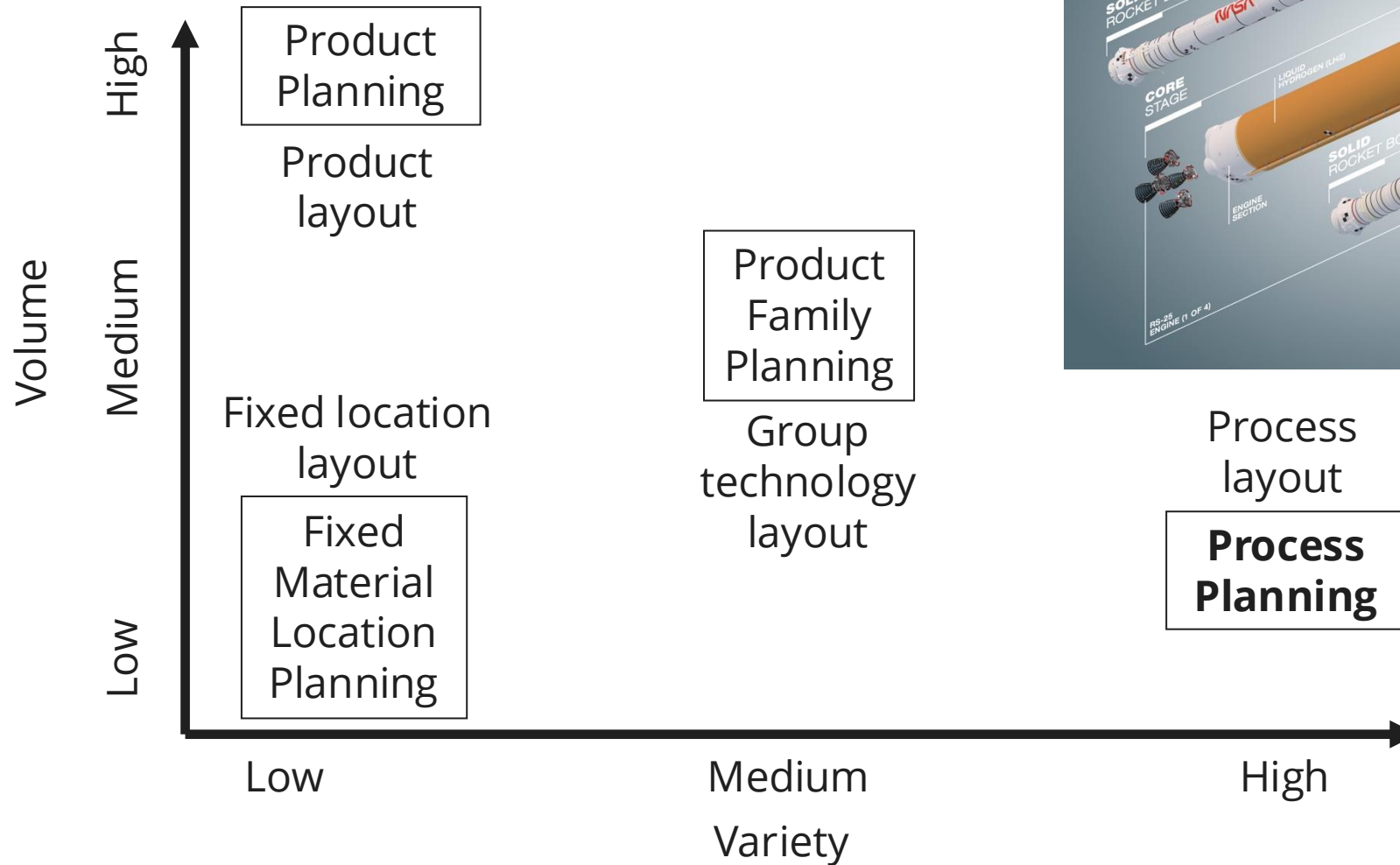
Advantages and Limitations of Layouts

- **Group layout**

Advantages	Limitations
1. By grouping products, higher machine utilization can result. 2. Smoother flow lines and shorter travel distances are expected than for process layouts. 3. Team atmosphere and job enlargement benefits often result. 4. Has some of the benefits of product layouts and process layouts; it is a compromise between the two. 5. Encourages consideration of general-purpose equipment.	1. General supervision is required. 2. Greater labor skills required for team members to be skilled on all operations. 3. Critically dependent on production control balancing the flows through the individual cells. 4. If flow is not balanced in each cell, buffers and work-in-process storage are required in the cell to eliminate the need for added material handling to and from the cell. 5. Has some of the disadvantages of product layouts and process layouts; it is a compromise between the two. 6. Decreased the opportunity to use special-purpose equipment.

Source: Facilities Planning (Tompkins, White, Bozer, Tanchoco) Table 3.2, pp.112-113

Departmental Planning



FTZ vs Bonded Warehouse:

Feature	FTZ Warehouse	Bonded Warehouse
Duty Deferral	Duties and taxes are deferred until goods enter the domestic market.	Duties are deferred until goods leave the warehouse for domestic distribution.
Manufacturing & Manipulation	Allows repacking, relabeling, light assembly, and sometimes full manufacturing.	Limited to basic storage and repacking.
Export Benefits	Ideal for goods that will be re-exported, potentially avoiding duties altogether.	Allows duty drawback if goods are re-exported.
Domestic Distribution	Requires customs clearance before goods enter the local market.	More suitable for domestic distribution as goods are already within the customs territory.
Setup Costs & Complexity	Higher setup costs and regulatory requirements.	Easier and less expensive to set up.
Storage Duration	Varies by country; often unlimited in the U.S.	Typically has a time limit (e.g., 5 years in the U.S. and EU, shorter in some Asian countries).

FTZ vs Bonded Warehouse:

FTZ vs. Bonded Warehouse United States

FTZ has full control of goods. 24 hours a day.	Surveillance 	Under continuous CBP surveillance.
Movement of goods is relatively unrestricted in and out of an FTZ. Consolidate weekly imports into one entry, lowering Merchandise Processing Fees due to capped fees.	Movement of Goods 	Movement of goods is limited in a bonded warehouse. Specific customs approval is required for each movement.
Allowed, with potential duty savings based on finished product.	Reforming Goods 	Not allowed.
Duty is reported when goods leave the FTZ.	Reporting 	Duty reported upon withdrawal of goods from the bonded warehouse for domestic use.
Customs formalities deferred until goods exit the FTZ for consumption.	Customs Clearance & Declaration 	Customs clearance required before goods enter or leave the bonded area.
No set limit.	Storage Limits 	Up to five years.
Various types of goods including raw materials, components, finished goods, etc.	Storage Goods Category 	General merchandise, goods subject to duties, taxes, or other restrictions.

FREE TRADE ZONE	VS	BONDED WAREHOUSE
Within US territory but outside US Customs territory.	Location 	Inside US territory and US Customs border.
Typically located near ports, airports, or international borders. US FTZs can be created within existing warehouses.	Customs Jurisdiction 	Secured facilities throughout the US.
Governed by the Foreign-Trade Zones Board under the U.S. Department of Commerce.	Competent Authority 	Managed by U.S. Customs and Border Protection (CBP).
Established under the Foreign Trade Zones Act of 1934.	Foundation of Law 	Governed by Title 19 of the U.S. Code and CBP regulations.
Goods considered outside U.S. customs territory.	In/Out Warehouse 	Imported goods awaiting customs clearance.
Foreign and domestic goods may be placed in a FTZ.	Permissible Cargo 	Only foreign goods may be placed in a bonded warehouse.
Duties and taxes are deferred, reduced, or eliminated on imported goods.	Tax Concession 	Duties and taxes are deferred until goods leave the bonded area for consumption.
Secure perimeter with CBP oversight, specific requirements vary by zone.	Security 	Security measures in place to prevent theft, loss, or unauthorized access.

Summary of Module 2

- Product design: which products will be produced and their detailed design
- Process design: how the product will be produced
- Schedule design: when and how much to produce
- Information gathering and representation
- Requirement calculation

